

Innovation Strategy after IPO: How AI Analytics Spurs Innovation after IPO

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Abstract

We examine the role of AI analytics in facilitating innovation in firms that have gone through IPO. Using patent data on over 1,000 publicly traded firms, we find that firms acquiring AI analytics capability post-IPO experience less of a decline in innovation quality compared to similar firms that have not acquired that capability. This effect is greater when only machine learning capabilities are considered. Moreover, we find this sustained rate of innovation is driven principally by the continued development of innovations that combine existing technologies into new ones—a form of innovation that is especially well supported by analytics. By examining three main mechanisms that hampered post-IPO innovation, we find that AI analytics can ameliorate the pressure to meet short-term financial goals and disclosure requirements. However, it has limited effect in addressing managerial incentives. For firms with long product cycles, the disclosure effect is reduced to a greater extent than it is for those with short cycles. Overall, our results show the importance of examining technology as a critical input factor in innovation. We show that the increased deployment of AI analytics may reduce some of the innovative penalties suffered by IPOs, and that investors and managers can potentially mitigate post-IPO reductions in innovative output by directing capital acquired in the IPO process to the acquisition of AI analytics capabilities.

Keywords: AI analytics, IPO, innovation quality, machine learning

Introduction

Early-stage firms are an important contributor to innovation in advanced economies, especially in critical sectors such as information technology and life sciences (Haltiwanger et al. 2013). It has been noted, however, that as early-stage firms develop, the quality of their innovative output declines after their initial public offering (IPO) (Aggarwal and Hsu 2013; Bernstein 2015; Wies and Moorman 2015). Explanations advanced for this reduction include increased short-term financial pressures, disclosure requirements that reduce the potential rents from innovation, and changes in employee incentives due to dilution of ownership. These effects are shown to be independent of accounting choices or the life cycle of the firm in which most-IPO firms may choose between capitalizing on their existing innovations or developing new ones (Asker et al. 2014). Individually or collectively, these reasons may encourage newly public firms to focus on the production of lower risk, incremental innovations that are more likely to be built upon their existing stock of knowledge and discourage pursuing riskier novel or breakthrough innovation.

Firms recognizing the potential for innovation decline after going public have responded in a variety of ways, most of which have been costly and at times, ineffective. For instance, Dell reverted to private ownership by completing a leveraged buyout in 2013. It did so to focus on long-term innovation made possible by being freed from meeting short-term financial targets demanded of public companies (Devaney 2013). To cite another example, Ikea pioneered selling affordable furniture and is one of the most innovative firms in the world. It has chosen to remain private to maintain a long view of innovation (Tronvoll et al. 2019). Public firms often acquire early-stage companies to improve their own innovation portfolio but this strategy is both costly and challenging due to cultural differences between mature and early stage firms combined with the difficulties of retaining key personnel after the acquisition (see e.g., Nahavandi and Malekzadeh (1988) and Aggarwal and Hsu (2013)). Furthermore, an acquired firm may soon experience a similar innovation decline once it is part of a publicly traded firm, and thus the relief it provides from the innovation decline to the acquiring firm proves temporary.

While the innovation decline after IPO has been well documented, limited research offers effective solutions to this problem. Firms can stem the decline in innovation using financially-oriented strategies such as leveraged buyouts (LBO) or mergers and acquisitions (M&A), but the success of these strategies is not certain. Such strategies are often expensive, risky and entail significant organizational change. Identifying less costly ways for firms to maintain their innovative momentum, especially in the activities common to private firms, can be immensely valuable. Recent advances in analytics technology, enabled by new machine learning capabilities and increased digitization efforts, could potentially mitigate some of the innovation decline faced by public firms without requiring complex financial restructuring. Collectively, we refer to analytics related to data analysis and machine learning as *AI analytics*¹ which can include both advanced machine learning methods as well as other data analytics skills, recognizing such methods not only require advanced machine learning skills but also data analysis skills to collect, process, analyze, and interpret data. In particular, recent work has shown that AI analytics capabilities can facilitate the process of searching, aggregating, and mining diverse knowledge which enables new combinations of technologies from prior art (Wu et al. 2020). By accelerating the rate of creating new combinations in innovation, AI analytics could potentially mitigate some of the innovation decline experienced by firms that have recently gone public.

This paper examines whether a) acquiring AI analytics capabilities reverses the decline in innovation quality following an IPO, b) whether this improvement (if present) is consistent with improving innovation through new combinations of existing technologies, and c) to what extent AI analytics can mitigate innovation decline by addressing the three main factors that contribute to the decline that have been identified in the literature (short-termism, disclosure requirements, and managerial incentives). To the best of our knowledge, our paper is among the first to examine how technologies can be used (as opposed to complex financial instruments) to mitigate the innovation decline after IPO and the underlying mechanisms that explain the mitigation.

¹ <https://www.cognizant.com/us/en/glossary/ai-analytics>

We begin by collecting data on firms that have done an IPO and have filed patents from 1988 to 2013 and replicate previous findings on the adverse relationship between completing an IPO and firm innovation. We then extend our data to 2019 to examine the effect from the recent advances in machine learning. We show that the greatest reductions in innovation quality result from a reduction in the development of technologies that involve the discovery of new combinations of existing technologies. Our findings also show that firms that increase their use of AI analytics can offset a substantial reduction in the type of innovation that involves a new combination of existing technologies. However, we observed a negligible and at times negative effect of AI analytics in fostering radically new innovation—that is, entirely novel technology domains. Examining the three mechanisms that have been attributed to the post-innovation decline in innovation—short-termism, disclosure requirements, and managerial incentives—we find that AI analytics can help mitigate the effect of short-termism and disclosures but does not counter the negative impact of managerial incentives on innovation. These results are robust using various methods (difference-in-differences, firm-level fixed effects, instrumental variables, Heckman two-stage estimation, and matching estimators) to address potential selection and reverse causality bias issues of our sample. We also observe that post-IPO firms are using IPO resources to ramp up hiring of employees with AI analytics skills within the first few years after their IPO, a fact that supports the potential of using AI analytics to boost innovation post-IPO. Collectively, our results suggest that ramping up AI capabilities acquired through hiring AI analytics skills can provide a mechanism that is likely to be much less disruptive than alternatives (e.g., financial restructuring) in fostering innovation. This is especially true for the types of innovation that AI analytics are well-positioned to support: new combinations of existing technologies. However, for post-IPO firms, AI analytics has a negligible role in fostering radically new innovation.

IPO and Innovation Outcomes

The effect of an IPO on firm innovation and productivity has received renewed attention in recent years. Despite (or possibly because of) having access to capital from the sale of their stock in the public market, it has been documented that firms often experience negative effects on operational effectiveness,

sales growth, and overall productivity following their IPO (Chemmanur et al. 2009; Jain and Kini 1994). Prior research suggests that the decline in performance post-IPO is not due to accounting choices or to public firms being in different life-cycle stages (e.g., being maturing firms in stable industries) but is related to firms becoming less likely to respond to market opportunities after going public.

What is most striking about these findings is that there are numerous reasons why an IPO should facilitate rather than curb innovation. For instance, a reduction in financial pressures created by newly raised capital should increase incentives to engage in risky innovation (reducing or eliminating the bankruptcy risk from failed large-scale projects); this does not appear to be empirically supported (Hall and Lerner 2010). Similarly, the reputational effect of a successful IPO can also engender many intangible benefits, including making it easier to acquire talented and innovative employees, strengthening bargaining power with suppliers, and signaling quality to consumers. Theoretically, these factors should provide more freedom for firms to experiment and to take on riskier innovation projects (Cyert and March 1963; Wies and Moorman 2015).

The current literature offers three main explanations for this somewhat counterintuitive finding that innovation drops rather than rises. First, IPO events may force firms to focus on short-term financial goals, discouraging risky investments and causing firms to target innovations that possess high-probability, near-term payoffs (Hall and Lerner 2010). This is compounded by the fact that it is often difficult for external investors to evaluate long-term innovations in progress that would require deep subject expertise about the innovation (Ferreira et al. 2012) thus preventing innovative firms from being suitably rewarded for these investments prior to the creation of observable financial benefits.

Second, publicly traded firms also face disclosure requirements from regulators, legislators, and other actors in the market. These include everything from mandatory stock exchange disclosure rules to investors and analysts demanding information required to properly value the firm. These types of disclosures reduce the potential rents from novel innovation (Brau and Fawcett 2006). Added to this is another force driving companies away from long-term risky innovation: imitation, which is less costly, faster and carries lower risk than innovation (Teece 1986) for both the focal firm and its competitors. The

combination of public scrutiny and potential leaks of sensitive innovative discoveries and processes to competitors could further reduce incentives for a firm to innovate, especially on risky long-term projects where risk of discovery rises over time.

Finally, the incentives of the innovators can also change after an IPO. Faced with dilution of their ownership stake, inventors may choose to avoid risky projects when the reward is now shared with a much larger group. Post-IPO firms often have difficulties in retaining key original innovators because these innovators may prefer a private firm where they can receive a larger share of the returns for their efforts (Aggarwal and Hsu 2013; Bernstein 2015). Even for inventors who stay after an IPO, retention can be a problem as new opportunities arise (e.g., spinoffs) that enable them to pursue innovation without the constraints of diluted ownership (Bernstein 2015). Furthermore, as the company grows, it is harder for owners to monitor managerial performance to distinguish between poor luck and poor managerial skills for long-term or risky activities, creating further disincentives for innovation (Aghion et al. 2013).

Thus, there are strong incentives for post-IPO firms to invest in incremental innovation as opposed to long-term and novel innovations. Innovators appear to act on these incentives. Bernstein (2015) and Aggarwal and Hsu (2013) use patent data to show that post-IPO firms did not necessarily reduce the overall number of patent applications, but that these patents have less impact than earlier patents as measured by citation metrics. Similarly, Wies and Moorman (2015) use product categories in the consumer product industry to show that publicly traded firms tend to invest in incremental innovation projects that are closely related to their own knowledge, as opposed to exploring radically new innovation. Some public firms try to mitigate the lack of internal innovation by acquiring other start-ups (Bernstein 2015) for their knowledge assets and human capital. Alternatively, some firms have made deliberate efforts to maintain their innovative position, incurring substantial real or opportunity costs, such as remaining private or engaging in costly financial restructuring to convert from public entities into private ones. Therefore, identifying a less costly way to sustain their innovative momentum can be immensely valuable.

AI Analytics and Innovation

The recent rise of AI analytics may provide a mechanism for alleviating the post-IPO innovation decline. This is in part due to the digitization of various social, economic, and individual behaviors that provide the necessary inputs for the application of large-scale data analytics. Through advances in artificial intelligence, many firms can mine large-scale data to improve product quality, customer service, pricing and product design, and business processes (Cockburn et al. 2018; Liu et al. 2023). With the advances in machine learning, AI analytics has since moved beyond incremental process improvement and can quickly detect patterns and unobserved linkages in divergent data sources to generate new insights and innovation. By keeping track of what is already known inside the firm and identifying and integrating external knowledge as it becomes available, AI analytics has accelerated the rate of innovation. The primary mechanism for AI analytics to create new innovation is through combining existing knowledge in new ways or by enabling disparate ideas to be combined to solve problems in a new domain (Wu et al. 2020). For example, Autodesk collaborated with engineers and car drivers to deploy numerous, low-cost sensors on racing cars and to collect vast amounts of new data on stresses, strains, temperature, displacement, and all other factors that might affect the performance of the chassis. Combining the new sensor data with engineering design principles, Autodesk deployed machine learning algorithms to generate an entirely new chassis design that is asymmetrical, reflecting the fact that track cars turn left more than right (McAfee and Brynjolfsson 2017). While it would have been possible for humans to find a similar solution, machine learning vastly shortened the timeline. Similarly, many venture activities in the pharmaceutical industry have focused on using analytics to find new drug targets and chemical compounds that could treat them (Lou and Wu 2021). It is increasingly recognized that machine learning algorithms can offset their disadvantage relative to human decision makers by operating at a faster rate, larger scale, and by considering more data and trial solutions than what would be possible without modern computing and data acquisition capability (Stokes et al. 2020).

AI Analytics and post-IPO Innovation

Thus, the ability of AI analytics to track, process, analyze, and transform existing knowledge across divergent sources to detect patterns, find useful insights, and support decision making should help firms innovate by finding new ways to combine existing information and reuse existing technologies (Wu et al. 2020). While AI analytics can help all firms innovate, post-IPO firms may derive greater benefit than others for two reasons. First, the cash infusion from an IPO allows firms to invest in AI analytics skills and technologies. Second, post-IPO firms face an innovation decline in the very types of innovation—new combinations—that AI analytics is designed to support; because this decline is in contrast to what their private counterparts face, post-IPO firms stand to derive greater benefit from analytics as a remedy.

The use of AI analytics can become feasible for newly public firms that acquire sufficient capital through their IPOs that permit them to acquire expensive analytics talent that was previously out of reach.² Even though this is costly, changing the skill mix of existing staff by adding new hires is likely to be less disruptive and considerably less expensive than engaging in alternatives like financial restructuring (e.g. such as LBOs), especially since these efforts can also entail significant organizational change and risk (Aggarwal and Hsu 2013; Nahavandi and Malekzadeh 1988).

Once the cash infusion supports investment in analytics, AI analytics can potentially alleviate some of the factors that contributed to the innovation decline after the IPO. While AI analytics cannot address all organizational and incentive factors that post-IPO firms face in maintaining momentum in long-term innovation, it can alleviate some of the disincentives by accelerating discoveries through exploiting hidden patterns and linkages across diverse data. Specifically, AI analytics can potentially alleviate some of the effects from the three main factors—short-termism, disclosures, and managerial incentives—that have discouraged post-IPO innovation.

² <https://www.forbes.com/sites/gilpress/2018/05/04/the-salaries-of-data-scientists-remain-steady-but-still-sexy-with-ai-on-the-horizon/#333ae4d9733d>

First, AI analytics can accelerate innovation, reducing problems associated with short-term financial pressure, and can broaden the search space which, in turn, diversifies risks and improves resource allocation by eliminating dead ends more quickly. By reducing some of the risks associated with long-term innovation, it can help relieve some of the pressure associated with meeting quarterly earnings goals.

The same acceleration of innovation can also reduce the disclosure pressure, the second factor that contributes to the innovation decline after IPO. Financial disclosures not only reveal financial investments, but they also often discuss business risks, R&D activities, and firm strategy. Shorter innovation cycles may enable intellectual property protection to be obtained before disclosure requirements enable imitation by competitors. Furthermore, the increased rate of innovation as well as the opacity of many types of machine learning or other analytics activities may reduce the rent dissipation and competitive spillovers of public disclosure. Often, the key to successfully using machine learning models lies in tuning millions or even billions of parameters which are context specific and thus cannot be readily inferred by outside observers. In addition, machine learning models are not often interpretable by competitors seeking to learn exactly how the algorithm arrives at a given result. If analytics-driven innovation is harder to reverse engineer, it may reduce imitation, perhaps especially important for technologies where patents provide limited protection.

Finally, managerial career concerns for taking on risky projects can be somewhat reduced when the resource commitment to any particular idea is lower and the time to identify unpromising ideas is shortened. However, the technology may have a limited ability to affect an innovator's incentives because certain innovators would prefer to work in a private firm to gain a greater share of the rent from their innovation. Furthermore, some innovators may prefer the culture of smaller startups as opposed to that of publicly traded firms. It is unlikely that any technologies, including AI analytics, can effectively bridge these cultural gaps. Lastly, some innovators may also want to take on projects that require radically new technology, the type of innovation that analytics is not well-suited to support.

Thus, collectively, we hypothesize that post-IPO firms can use AI analytics to improve innovation because AI analytics can counterbalance the negative effects of meeting short-term financial goals and disclosure requirements. As AI analytics is particularly effective at recombining existing knowledge into new knowledge, we expect analytics to help post-IPO firms improve their innovation qualities in these aspects. Simply stated, we hypothesize that: *AI analytics facilitates innovation by post-IPO firms by combining existing technologies in new ways.*

Data and Measurement

To understand how AI analytics can mitigate the decline in innovation quality for post-IPO firms, we measure (1) the level of innovation activity before and after a firm's IPO, (2) the AI analytics skills in a firm before and after an IPO, (3) the channels by which AI analytics can mitigate the innovation decline after the IPO, and (4) the financial performance before and after the IPO.

Innovation Quality

Patent analysis has been extensively adopted as a tool in economics and management research to measure innovation and technological progress. We combine the latest version of the NBER patent data file that contains detailed firm-level patent data through 2006 with the more recent patent data from the USPTO (2007-2019) that we process using an identical approach (Hall et al. 2001).³ To measure innovation outcomes at firms, we start with a firm's patent count, calculated as the number of patents filed by the firm in a given year (Aggarwal and Hsu 2013; Saunders 2011) that are eventually granted by the USPTO. We use the application year of the patent rather than the issue year because the filing date more closely approximates when the firm produced and started using the innovation proposed in the patent application.

Patents are not equally valuable, and the existing literature articulates a concept of patent or innovation "quality" to capture these differences. The measurement of this concept varies. To measure

³ <https://sites.google.com/site/patentdatapoint/Home> covers the 1976-2006 patents. We also use other data sources, including the earlier version of the data available at <http://www.nber.org/patents/>, USPTO, and patents stored on Google, for patents before 1976.

innovation quality, we calculate two widely used metrics (Hall et al. 2001). First, we measure *originality*, or *technological diversity* (Kaplan and Vakili 2015; Wu et al. 2020) computed from backward citations (that is, using citations to the prior art in the focal patent to measure the diversity in the set of knowledge that a patent draws from). Second, we measure *generality* based on the number of forward citations (that is, using future patents that cite the focal patent to measure how widely this patent is cited by others). These metrics have been used extensively in the literature on R&D (Hall et al. 2001; Trajtenberg et al. 1997) and for quantifying innovation quality in the corporate finance literature (Brav et al. 2018; Lerner et al. 2011). We further refine our measures of innovation quality by constructing three additional measures to capture different types of innovation in each patent: (1) reuse, (2) new combinations, and (3) radically new inventions. Specifically, we use patent subclasses of the focal patent and those that it cites which provide a very fine-grained measurement of a patent's technology domain.⁴ A patent qualifies as reuse if it has the same technology or the same combination of technologies as an existing patent. A patent is a new combination if it is the first to combine an existing set of technologies in a new way (that is, all the prior technologies exist, but not in that combination). A patent is a radically new invention if it is the first patent under a specific technology domain, and hence the first innovation to create the new technology. The three innovation types above (reuse, new combination, and radically new invention) can be calculated on two levels of scope, relative to the firm (local) or to the market as a whole (global) (Lin et al. 2021). For instance, a patent is a radically new invention for a firm (local) if it is the first time that firm has created a patent in that technology domain. We also consider the scenario where the patent is the first ever to be in a technology domain, in which case the patent would be considered novel globally

⁴ In the US Patent Classification System (USPC), patents are classified at two levels—patent class and patent subclass. Our analysis is mainly focused on the more granular subclass level, and a simple term “technology” or “technology domain” is used to refer to an individual patent subclass for expositional ease (Akcigit et al. 2013). The USPTO defined 475 classes and 165,000 patent subclasses (<https://www.uspto.gov/web/offices/ac/ido/oeip/taf/cbcby.htm>). As robustness checks, we also use classifications of patents based on their Cooperative Patent Classification (CPC) subgroups. For those patents without USPC codes but with CPC codes, we obtain the correspondence between USPC subclasses and most relevant CPC subgroups (for example, <https://www.uspto.gov/web/patents/classification/cpc/html/us706tocpc.html>). Our estimation results are consistent across different ways of patent classifications.

(across all firms), not just locally (within the firm).⁵ The three types of innovation and the two levels of scope yield six types of innovation. However, since these metrics are highly correlated, we focus on reuse and new combinations at the firm (local) level and on novel-tech at both the local (*l*) and global (*g*) levels. The final metrics are simply the sum of patents (*m*) of that type for each firm in each year.

$$\text{Reuse}_{[l]} = \sum_{j=1}^m 1(\text{existing technology domain or combination}_j)_{[l]} \quad (1)$$

$$\text{New combination}_{[l]} = \sum_{j=1}^m 1(\text{new combination}_j)_{[l]} \quad (2)$$

$$\text{Radically new invention}_{[l,g]} = \sum_{j=1}^m 1(\text{first of a technology domain}_j)_{[l,g]} \quad (3)$$

AI Analytics

We measure a firm's AI analytics capability (*Analytics*) by calculating the total number of employees who possess relevant AI analytics skills. This can include generally data analytics skills as well as advanced machine learning skills. We use two different data sources to comprehensively measure the overall AI analytics capabilities in firms from 1988 to 2013. First, we use 6 million individual resumes that depict the career trajectory of these sampled employees from 1988 to 2007. This dataset is similar to other large-scale datasets of resumes exploited in prior IT value literature (Tambe and Hitt 2012; Wu et al. 2018; Wu et al. 2019). It provides a detailed timestamp of employment history including detailed job descriptions and has good coverage of both public and private firms (Tambe and Hitt 2012). However, in our analysis, we also create machine learning (*ML*) skill variable to capture the most recent advances in machine learning (described in the next section).⁶ To obtain more recent data about AI analytics capabilities in firms, we use job review data from 2008 to 2013 from a large online job review platform that contains job titles of employees who provided the review.

We first develop a taxonomy of AI analytics skills, which includes basic skills such as data analysis as well as advanced machine learning skills. We choose to be encompassing in our AI analytics

⁵ The local vs. global distinction can also be applied to new-combination and reuse types of innovation. We find the values of the two types of innovation are highly correlated with their local types of innovation, and they yield consistent estimation results.

⁶ We follow the recent literature on the economic value of AI to distinguish machine learning from other types of AI investment (Cockburn et al. 2018; Lou and Wu 2021; WIPO 2019).

measurement because a successful execution of advanced machine learning algorithms requires data analysis skills. We then separately measure AI analytics skills that pertain to machine learning only.

In measuring AI analytics, our approach uses matches on general keywords such as “data analytics”, “artificial intelligence” as well as those that denote analytics concepts such as “regression modeling,” “machine learning,” and “Hadoop” that represent algorithms or technologies associated with AI analytics (Table A1 in Appendix A). We also rely on the classification used by Saunders and Tambe (2015), Cockburn et al. (2018) and a widely accepted computing classification system from the Association of Computing Machinery Computing Classification System that accounts for the dynamic change of AI technologies (Lou and Wu 2021; WIPO 2019) to refine and validate our keywords. Using a few variants of these keywords does not lead to significant differences in our measurement. We corroborate this measure using the job titles from the Occupational Information Network (O*NET)⁷ (a government database that maps jobs to the skills these jobs require) and find the two measures and the outcome of employing these measures in our models to be consistent. Once the relevant skills are identified, we examine when the skill was in use and the employer where it was used. To measure the AI analytics capability of the firm, we aggregate all the employees within the firm in a particular year who were identified as having AI analytics skills. Because not all employees can be observed in our data, we estimate the number of employees with AI analytics skills by using a firm-level sampling rate computed by $\theta_i' = \frac{x_i}{L_i}$, where x_i is the number of employees across all occupations in our sample for firm i , and L_i is the total number of employees for firm i as provided by Compustat and other proprietary databases including Bureau van Dijk Orbis. In this estimation procedure, we assume that the observed employees

⁷ To broadly capture AI analytics associated with data analysis and machine learning, we search for the word “data” in the Occupation Search box provided by <https://www.onetonline.org/>, and obtain a list of job titles ranked according to how well these job titles matched the keyword “data,” according to the matching process in <https://www.onetonline.org/help/online/search#keyword>. A total of 576 primary data-related job titles are generated, with detailed reports on the requisite technology skills and general skills. An additional 212 alternate occupation titles are found from the primary job titles. For example, the occupation “Database Administrator” has several alternate job titles, such as data engineer, data miner and data center manager. We examine these job titles, and count a person as having AI analytics skills if any of the skills in that job title is related to data analysis as well as machine learning skills.

are sampled from the underlying population of all workers in the firm. We also assume that firm and occupation-specific factors regarding the likelihood of posting career trajectories or job titles are uncorrelated. In earlier research the same method was used to capture data skills and IT skills (Tambe and Hitt 2012; Wu et al. 2019; Wu et al. 2020).

To examine a firm's investment in AI analytics skills in a more recent period from 2008 to 2013, we gather more than 3.7 million job reviews with metadata including job title, employer, and length of employment; the data are acquired from a leading online job review platform. These online reviews are written by employees of the firms. Because they are less fine-grained than the resumes that have detailed job descriptions including skills, we identify AI analytics skills using job titles in each review and then aggregate them at the firm level. We rely on O*NET and perform an occupation search on that platform to find job titles that are related to AI analytics. Each employee is identified as having AI analytics skills if any of the skills listed under their job titles are related to AI analytics.⁸ We also complement the O*NET-based classification by using a simple search of AI analytics-related keywords in job titles ("artificial intelligence", "data", "machine learning", "statistics", "business analytics"). As with the resume data, we identify the length of various employment spells at a particular employer, aggregate employees with related AI analytics skills by length of time for each employer and then adjust by a firm-specific sampling rate in each year.

To reconcile the potential differences between the two data sources, we compute appropriate scales to convert measures from one data source to the other and capture firm-specific scaling ratios by using the common periods between the two data sources (Figure A1 in Appendix A). A dummy variable for a data source can be used to control for the potential heterogeneity arising from how AI analytics is measured, but the dummy is subsumed by the yearly dummies as the data sources do not overlap in time in the final panel data. In addition, Chow tests ensure the merging of two data sources is valid for our

⁸ Wu et al. (2020) use weighted skills based on job titles as some job titles list more AI analytics-related skills, which do not change our main results.

empirical models. Lastly, we adopt a single job title classification based on O*NET; our results are very similar and corroborate earlier studies that use similar methods (Wu et al. 2020).

Machine Learning Skills

While AI skills and general data analytics skills are tightly linked, we also create a separate skills measure that is only related to machine learning to reflect recent advances in machine learning. It is a subset of the skills in the larger AI analytics skills. We use job-posting data to create measures about personnel skills related to machine learning. The data are obtained from a leading analytics company that actively tracks the changing workforce and labor market over time. The job postings are collected from over 40,000 online job boards and company websites from 2010 to 2019. Similar to the methods developed in the recent literature on the economic value of AI (Alekseeva et al. 2021; Babina et al. 2022; Lou and Wu 2021), we examine the skill requirements in each posting and search for a similar set of words commonly related to the knowledge of machine learning and development of machine learning software (Table D1 in Appendix D). If any of the skills listed in the job posting is related to machine learning, we treat the posting as requiring machine learning skills. We aggregate the number of machine learning-related job postings for each firm in each year and use it to estimate the total machine learning skills.⁹ Thus, it can capture the latent demand for machine learning skills in firms, which is critical to engaging in AI-related tasks and using machine learning tools to innovate. Then, we estimate the role of machine learning skills in post-IPO innovation in the recent period after 2010.

Three Channels by which AI Analytics Mitigates Post-IPO Innovation Decline

Short-termism: We create a variable (*Myopic*) to indicate whether a firm engages in management that focuses on the short term. Managers in a firm overemphasize activities with immediate payoffs to achieve short-term goals at the expense of investments in pursuing more distant payoffs (Mizik 2010; Stein 1989). In particular, managerial myopia manifests in an underinvestment in long-term assets

⁹ Babina et al. (2022) use job postings and find consistent patterns across various measures of AI skills.

such as marketing and R&D spending with the intention to create earnings for the short term (Chakravarty and Grewal 2011; Wies and Moorman 2015).

Following the methods developed in Mizik (2010), we identify the potential myopic firms: firms simultaneously reporting greater-than-normal profits, lower than-normal marketing intensity, and lower-than-normal R&D intensity in a given year. We use R&D expenditures divided by total assets as the measure of R&D intensity; for marketing intensity we use selling, general, and administrative expenditures minus R&D expenditures and divide the difference by total assets. Firm profitability is measured as operating income before depreciation divided by total assets. These firm-level financial and accounting measures are collected from Compustat and Bureau van Dijk Orbis databases. Then we employ autoregressive panel data forecast models to calculate normal levels of firm profitability. Specifically, to derive the “normal” or expected level of firm profitability, we regress the current period of firm profitability ($Profitability_{it}$) on its lagged one-year variable ($Profitability_{it-1}$) with year- and industry-fixed effects. The forecast value, $\widehat{Profitability}_{it|it-1}$, can be derived from the model and reflects the normal or expected level of profitability. Thus, we use the forecast errors (i.e., $Profitability_{it} - \widehat{Profitability}_{it|it-1}$) as the deviation from the normal level in each year. Similarly, we obtain the deviation from the normal level of marketing intensity ($Marketing_{it} - \widehat{Marketing}_{it|it-1}$) and R&D intensity ($R\&D_{it} - \widehat{R\&D}_{it|it-1}$). Thus, using these deviation values, we classify the firms with $(Profitability_{it} - \widehat{Profitability}_{it|it-1}) > 0$, $(Marketing_{it} - \widehat{Marketing}_{it|it-1}) < 0$, and $(R\&D_{it} - \widehat{R\&D}_{it|it-1}) < 0$, as myopic firms (*Myopic*) that overemphasize short-term goals in a year.

Information Disclosure: Firm disclosure of information to the public can hurt the quality of its own innovation development (Chen et al. 2021). We use two methods to approximate the disclosure. First, we measure whether firms disclose earnings forecasts which can help investors properly evaluate firms. Following the measurement developed in Bertomeu et al. (2022), we use a comprehensive set of managerial guidance data from I/B/E/S tracking management earnings forecasts to create an indicator variable (*Disclosure of earnings forecast*) about whether a firm discloses earnings forecasts in a year.

Management earnings forecast disclosure is one of the most common types of information disclosure measures (Chen et al. 2015). Managers in a firm can forecast its earnings and release them for different time horizons. These disclosures are not simply financial disclosures, but they are also relevant to technology investments, product pipelines, and firm strategy. The forecast release allows outsiders to assess managers' ability to make investment decisions (e.g., in new technologies and product pipelines), adjust production plans and anticipate changes in the firm's economic environment (Trueman 1986). Thus, this measurement can dynamically uncover the disclosure of forward-looking statements made by a firm and provides insight into its future financial performance as well as future firm strategy and product development. We focus the measurement on the time period from 2004 to 2019, due to changes in regulations for firm disclosures in the US in 2000-2002 (i.e., Regulation Fair Disclosure (Reg FD) in 2000 and the Sarbanes-Oxley (SOX) Act in 2002) that have significantly altered both the incentives of corporate information disclosure and the collection of management forecasts (Bertomeu et al. 2022).¹⁰

Second, we create an alternative time-invariant measure of disclosure (*VC backed*) that indicates whether a firm is backed by venture capital because a firm backed by venture capital discloses more information to receive funding (Guo et al. 2004). We track the cumulative amount of VC funding received by the firm using datasets from Thomson Reuters SDC VentureXpert and Crunchbase.

Managerial Incentives: Another channel that can explain the post-IPO innovation decline is related to the shift in managerial incentives towards more incremental innovation: firms with nonentrenched managers, characterized by managers who make themselves less costly to replace, experience a significant decline in innovation quality after the IPO (Bernstein 2015; Holmström 1999). Nonentrenched managers are exposed to the risk of losing jobs, and thus have lower incentives to take the risks needed to create high-quality innovation and can be even averse to innovation.

¹⁰ Since August 2000, Regulation Fair Disclosure prohibits firms from disclosing important information to market analysts and certain shareholders. Thus, it closes a private communication channel, and facilitates public managerial forecasts. In addition, SOX (enacted on July 30, 2002) mandates a set of internal procedures to ensure accurate financial disclosure and enhance firms' management responsibilities.

We create a time-invariant variable (*CEO chaired*) that indicates whether a firm's CEO is also the chairman of the board after the firm goes public, which serves as a proxy for managerial entrenchment (Bernstein 2015; Shleifer and Vishny 1989). These CEOs are more likely to take on risks needed to create high quality innovation. We use data from BoardEx that collects profiles of senior executives from companies across various industry sectors (Fracassi and Tate 2012). We analyze their employment history including firms they worked for, years of employment, and roles to determine whether they are both CEO and chairman of the board.

Firm IPO and Other Financial Characteristics

We use several databases including Compustat, Bureau Van Dijk Orbis, and Thomson Reuters SDC New Issues to identify IPO events from 1988 to 2013 (Aggarwal and Hsu 2013; Bernstein 2015; Wies and Moorman 2015). Our key measure of the IPO event is an indicator variable (*Post IPO*), which is zero for the years before an IPO and 1 for the year of the IPO and subsequent years. We exclude financial firms with standard industry classification (SIC) codes between 6000 and 6999, as recommended in the IPO literature (e.g., Bernstein (2015), p. 1369 and Bajo et al. (2016), p. 384). We rely on Compustat, Crunchbase, and Bureau van Dijk Orbis databases to collect other financial characteristics that may affect IPO events; these databases are used in prior literature (e.g., Hitt et al. 2021; Lou and Wu 2021; Wu et al. 2019). They provide rich information about public and private firms across industries during our observation periods. We use the logarithm of a firm's total assets to control for firm size and use R&D expenditures that may affect a firm's innovation capabilities (Bernstein 2015). We also calculate firm age to account for firm-level quality and life-cycle considerations (Aggarwal and Hsu 2013; Loughran and Ritter 2004; Wies and Moorman 2015).

The final panel data consists of 1,471 firms from 1988 to 2013. It involves two groups of firms: 1,080 changeover firms undergoing an IPO, and 391 firms remaining private throughout the observation period. We also extend the sample period to 2019 to examine the role of recent advances in AI analytics, enabled by new machine learning capabilities. Table 1 reports summary statistics and correlations of

variables used in our analyses.¹¹ Table 2 shows the industry breakdown using two-digit North American Industry Classification System (NAICS) codes for the firms in our sample. Multiple industries are covered reasonably well by our sample. We note that the majority of the firms are in the manufacturing industry, consistent with sample coverage in prior IT value literature (Bardhan et al. 2013; Chung et al. 2019; Wu et al. 2018).

Empirical Approach and Identification

To examine the pattern of corporate innovation activities around an IPO event and how a firm's AI analytics capability affects these activities, we extend prior empirical strategies that are used to study the effect of an IPO on firm performance (Chemmanur et al. 2009; Wies and Moorman 2015) by incorporating AI analytics capabilities as well as their interaction with the IPO event. In our estimation model, we relate various corporate innovative outputs to a firm's ownership status (*Post IPO*), AI analytics capability (*Analytics*), controlling for years (γ_t) which can address temporal innovation shocks and fluctuations in market movements that may impact IPO decisions (Chemmanur et al. 2009). Firm-fixed effects are also used in all models (γ_i) to account for ex-ante differences in innovation and the use of AI analytics (thus, we can interpret our results as the marginal effect of changes in AI analytics).¹² We control for firm age, total assets, R&D expenses, and number of patents. Our primary estimating equation is thus:

$$\ln(\text{Innovation Quality})_{it} = \beta_0 + \beta_1 \text{Post IPO}_{it} + \beta_2 \ln(\text{Analytics})_{it} + \beta_3 \ln(\text{Analytics})_{it} \times \text{Post IPO}_{it} + \gamma_t + \gamma_i + \text{Controls} + \epsilon_{it} \quad (4)$$

The focus of our analysis, β_3 , measures the effect of AI analytics skills on innovation quality after a firm has gone through an IPO. Here we measure innovation quality using a variety of metrics. Specifically, we first examine three metrics (originality, generality and forward citations within four years of the year in which the patent was filed) that are traditional R&D metrics used extensively in the prior

¹¹ The variable, *Disclosure of earnings forecast*, is not included in Table 1 as it's only available after 2004. It has a mean of 0.35, and a standard deviation of 0.48.

¹² Using a specification with one-period or longer differences yields similar results.

literature; we replicate earlier findings using our data. Then we use more detailed innovation metrics, reuse, new-combination, radically new invention within the scope of firms' innovation processes, as well as radically new inventions in the global scope that allow us to distinguish precisely the types of innovation that analytics can help post-IPO firms to create (Akcigit et al. 2013; Wu et al. 2020). We adopt similar regression specifications for the mechanism tests, replacing the variable about a firm's ownership status (*Post IPO*) with the variables about the three channels by which AI analytics may reduce the innovation decline (i.e., *Myopic*, *Disclosure of earnings forecast*, *VC backed*, *CEO chaired*).

There are several sources of endogeneity associated with Equation 4 as the IPO decision is discretionary, conditional on a variety of firm and market factors. The decision to invest in AI analytics is also subject to similar issues in reverse causality and selection biases such as the "free cash flow" hypothesis that firms with greater resources could be more likely to choose to invest in analytics than firms with fewer resources could.

To address potential selection biases on the choice of going through an IPO, we follow a recommended framework in the existing literature that uses matching estimators, specifically propensity score matching (PSM), to find a comparable sample of firms that are similar to the IPO firms but didn't go through an IPO (Aggarwal and Hsu 2013; Bernstein 2015; Wies and Moorman 2015) including those that filed for an IPO but did not complete it.¹³ The matching is conducted on a set of observed firm characteristics including firm age, firm size (we use a firm's total assets as a proxy), R&D expenditure,

¹³ While the effect of an IPO event on firm innovation has already been widely used in prior literature and is not the main focus in our study, there could be some concern about whether the parallel-trends assumption holds in the sample—that is, in the absence of treatment (firms' IPO event in our setting), the difference between the "treatment" group (firms going public), and the "control" group (firms remaining private) stays constant over time. To address this concern, we examine whether there is any differential pre-trend in the outcomes of interest between firms remaining private and those going public, using leads and lags analysis (Autor 2003; Sun et al. 2020). Specifically, we create four dummy indicator variables that are equal to 1 only in their corresponding year: 3 years before IPO and backward (*IPO Year_{t-3 backward}*), 2 years before IPO (*IPO Year_{t-2}*), 1 year before IPO (*IPO Year_{t-1}*), and the year of adoption and forward (*IPO Year_{t forward}*). In other words, for a firm *i* in year *t*, only one of these indicators is equal to 1. We undertake estimations using Equation 4 by replacing the variable *Post IPO* with these indicators. We consider the indicator *IPO Year_{t-3 backward}* as the baseline (i.e., omitted variable) in the estimation. As our main findings show patent originality and new-combination innovation type decline the most after IPO, we use them as dependent variables in the estimation. We find that the coefficient estimates for the indicator *IPO Year_{t-2}* and *IPO Year_{t-1}* are insignificant when our outcomes of interest are examined. The firms pick up the effect since the IPO year. Thus, we find no evidence of pre-trend differences that might have affected our findings.

patent count and VC financing (i.e., the cumulative amount of VC funding received by the firm). A matched sample is constructed to limit observations in the two groups to those that are as similar to each other as possible (Appendix C).¹⁴ Using PSM in combination with other parametric model estimations can substantially alleviate concerns from selection bias when making causal references (Gu 2022; Ho et al. 2007). We also use a Heckman two-stage estimation to further address the selection biases that relate to the choice to file for an IPO. The first-stage probit model predicts the probability of firms going public after controlling for the same set of observed firm characteristics as used in the PSM analysis.¹⁵ We then use the inverse Mills ratio (IMR) calculated from the first stage to correct for selection bias (Wooldridge 2010) in the second stage regression, as recommended in the literature (Wies and Moorman 2015).

To address the reverse causality of IPO firms choosing to pursue more incremental innovation, we identify firms having a lower probability of going public (from the probit model) that still complete an IPO filing and construct an indicator variable, unanticipated IPO (*UIPO*), for such firms. This indicator becomes 1 if the firm i unexpectedly goes public at time t and 0 otherwise. We then regress the one-period lag of innovation outcomes on this indicator variable. We find that this indicator does not predict innovation outcomes, suggesting that past corporate innovation decisions are not related to IPO decisions (Brau and Fawcett 2006). This corroborates findings from earlier work that uses the same method to show that reverse causality is not a serious concern in linking IPO events to innovation (Wies and Moorman 2015).

¹⁴ The matching is conducted on the two group of firms: the treatment group consists of firms going public and the control group consists of firms that remain private in our observation period. We use the nearest neighbor matching algorithm to identify the closest $k = 5$ firms in the control group for each treated IPO firms, with a caliper of 0.0001 (Liu and Bharadwaj 2020). The matching is performed based on the observed firm characteristics including firm age, total assets, R&D expense, patent count and VC financing. We examine the covariate balance before and after matching to evaluate the performance of matching (Table C1 in Appendix C). We find that the two groups are statistically different along each observed variable before matching. After matching, the differences in means between the two groups of firms become insignificant across the observed variables, suggesting that selection bias is meaningfully reduced. As robustness checks, we test the matched sample with closest $k = 3$ nearest neighbors and a caliper of 0.001, and we also use Coarsened Exact Matching (CEM) approach (Iacus et al. 2012). Our estimation results on the samples using different matching approaches are consistent (Appendix C).

¹⁵ Adding industry dummies for the prediction as suggested in the literature (Wies and Moorman 2015; Xiong and Bharadwaj 2011) yields consistent results.

Next, we use instrumental variables to address the potential reverse causality of AI analytics on innovation outcomes. It is possible that innovative firms with slack resources could recruit more employees with AI analytics skills which could be observationally equivalent to our preferred interpretation that analytics supports innovation. We use three sets of instrumental variables derived from a firm's hiring network (firms that hire from each other) to create instrumental variables for analytics skills: 1) the total number and fraction of employees with AI analytics skills in a firm's hiring network; 2) the adoption of an enterprise resource planning (ERP) system by neighboring firms (one-degree separated) in the hiring network and the total number and proportion of neighbors starting to use ERP systems; 3) the neighboring firms' adoption of a Human Capital Management (HCM) system in the hiring network and the total number and the proportion of neighbors starting to use HCM systems. These instrumental variables are similarly used to address endogeneity related to firm-level decisions to invest in analytics (Wu et al. 2020).¹⁶ Note that the instruments are not constructed from competitors in the same industry or firms in the same region. Our "similar" firms are neighbors in the hiring network (firms that hire from each other). They vary substantially in geography and industries, especially when considering hiring technical skills such as AI analytics skills (Wu et al. 2018). Accordingly, they are less subject to common industry shocks and competitive pressure as when using a competitor's characteristics to instrument one's own.¹⁷

¹⁶ The first instrumental variable is constructed based on the neighboring firm's existing talent with AI analytics skills. It measures the potential labor pool accessible to the firm through employee mobility, which reflects the cost of acquiring additional AI analytics talent. The second and third instrumental variables are derived from measures of the adoption of ERP systems broadly and the HCM module by firms in the same labor pool. These large enterprise-IT systems often involve the extensive application of analytics, and its use in a firm could create a shift in skill composition that can change the overall access to AI analytics skills for firms that share the same labor pool as the focal firm. As the ERP-adopting firms acquire employees with these skills during implementation and employees transfer these skills through job-hopping, the neighbor firms in the hiring network could also experience a change in access to analytics talent without going through any change on their own. The ERP measure is based on the presence of ERP-related keywords in employees' resumes. The HCM measure is based on the reported adoption of the HCM module in customer data from a large ERP vendor.

¹⁷ Firms that hire employees with similar skills may face similar labor-related risks and shocks, and exhibit a high co-movement of stock returns and accounting-based performance variables (Li 2017). Note that neighboring firms vary substantially in geography and industries and are not necessarily located in the same geographic regions and industry sectors as the focal firms that hire from them. Thus, the likelihood of the focal firms and their neighboring firms simultaneously hiring employees with similar skills becomes less probable, resulting in less similar labor-related shocks and a lower co-movement of their financial performance.

As a further check, we note that if there is indeed a reverse causality issue in which more innovative firms disproportionately use resources from an IPO to invest in analytics, we would expect that AI analytics and IPO to positively interact to improve innovation quality for all types of innovation outcomes. If we only observe the effect in some types of innovation but not in others, this could help rule out certain simple cases of selection effects such as the possibility that innovative firms generally choose to invest in analytics. This is what we find. Furthermore, we examine whether AI analytics can relieve the pressure from the three factors behind the innovation decline after IPO (short-termism, disclosure, incentives). If so, it is likely that AI analytics is responsible for mitigating the innovation decline. We have indeed shown evidence of these intermediate mechanisms.

Finally, if it is optimal for post-IPO firms to invest in analytics to spur innovation, we would expect firms to acquire more AI analytics capabilities post-IPO. To examine the dynamic change of a firm's demand in AI analytics around its IPO event, we adopt a similar regression framework to the one proposed by Bernstein (2015) to identify changes in firm capabilities in analytics. If investing in AI analytics capabilities can help firms innovate post-IPO, we would expect the firms to increase AI analytics investments over time. Instead of a single binary description of a firm's ownership status, we use $Event\ Year_{i,k}$ that is a dummy variable indicating k years before and after firm i 's IPO event. Analogous to Equation 4, we control for firm fixed effects (γ_i), year fixed effects (y_t) and time-varying variables of firm characteristics including firm age, total assets, R&D expenses, and patent count controls.

$$\ln(Analytics)_{it} = \beta_0 + \sum_{\substack{k=-3 \\ k \neq 0}}^{k=5} \beta_k IPO\ Year_{i,k} + y_t + \gamma_i + Controls + \epsilon_{it} \quad (5)$$

Overall, we believe that the combination of approaches using the instrumental variables, fixed effects models, Heckman two-stage estimation and matching estimators can reduce the concern about the potential biases in the estimation results from endogeneity problems. We acknowledge that it would be challenging to address all endogeneity concerns definitively when studying the economic impact in broad sectors of the economy. However, in addition to our various estimation methods, the use of multiple dependent variables (where effects are present some but not others), showing the effect of the three

mechanisms, and demonstrating increases in hiring AI employees after IPO should provide confidence that our results are likely to be causal.

Results

IPO and Innovation

First, we examine the relationship between corporate innovation quality and the public or private status of firm ownership by replicating earlier work using our data. Following Aggarwal and Hsu (2013), we examine the effect of an IPO on the aggregate patent characteristics at the firm level. Originality, generality, and forward citations within a 4-year period after the patent's application year all declined post IPO (Table 3).¹⁸ They are consistent with earlier studies that document a general decline in post-IPO innovation. In addition, we estimate the effects of an IPO on firm innovation using more fine-grained metrics of innovation that distinguish three different types of innovation (reuse, new combination, radically new) (see Equation 1-3). We find that the only innovations that decline significantly post IPO are new combinations of existing technologies, while other types of innovation remain steady.

AI Analytics Mitigates Innovation Decline

Next, we show how having AI analytics capability could moderate the innovation decline post-IPO (Table 4) using Equation 4. We aggregate the patent characteristics of all patents produced in a firm in a particular year and control for the total patent count to account for the fact that firms with more patents can have mechanically higher scores on some patent metrics (e.g., total forward citations). We also control for a firm's general R&D expenses. In Columns 1-3, we show that going public is associated with a decrease of 6% in the overall originality score, a 5.1% decrease in the generality score, and a 14.3% decrease in forward citations.¹⁹ However, we find that AI analytics skills can mitigate some of the

¹⁸ Earlier work also uses patent-level analysis (Bernstein 2015, p. 1374) to measure the impact of an IPO on using the citation characteristics of each patent. Consistent with Bernstein (2015), we find that patent originality (based on backward citations), generality (based on forward citations) and forward citations themselves gradually decline starting from one year after a firm's IPO, as shown in Figure B1 of Appendix B. For example, compared to the year of IPO, by the fifth year after the IPO there are already eight fewer forward citations.

¹⁹ We follow Aggarwal and Hsu (2013) that use forward citations within four years of the application date. Results are directionally consistent if we consider other length of time forward.

declines. We find that all else being equal, on average, a 1% increase in a public firm's AI analytics is associated with 0.01% increase in patent originality. Equivalently, doubling AI analytics after going public is associated with about 0.7% increase in patent originality (Column 1). However, our analysis shows that analytics does not seem to have a statistically significant effect on improving generality or forward citations (Column 2-3) for post-IPO firms. This could be due to a lack of sufficient statistical power to estimate these effects precisely or it could be that citation-related metrics are too crude to detect changes in novelty (Boudreau et al. 2011).

To explore more precisely how AI analytics can reverse the innovation decline, we also estimate the effects of AI analytics on three different types of innovation (reuse, new combination, radically new) for post-IPO firms. While we observe the post-IPO decline only in innovations that are new combinations of existing technologies, we find that having AI analytics capabilities can significantly reduce the decline of this type of innovation following an IPO (Column 6). All else being equal, on average, a 1% increase in a firm's AI analytics after going public is associated with 0.01% increase in new-combination patents relative to the firm's own knowledge in a year. Equivalently, doubling AI analytics after going public is associated with generating approximately one additional new-combination patents (that are further granted) if a firm has a total of one hundred new-combination patents. This result suggests that AI analytics is particularly effective at integrating existing knowledge and can aid post-IPO firms in creating innovations that combine existing technologies into new ones.

However, we do not find similar effects of AI analytics on radically new innovations that create new technology classes in the patent classification system (Column 5) or those that reuse technologies within the firm (Column 7). Interestingly, when we examine the effect of AI analytics on radically new innovations that are new to the greater environment (across all firms), we find the reverse. Rather than mitigating the innovation decline, we find that analytics has a negative and statistically significant effect on generating this type of innovation (Column 4), suggesting the limitations of using AI analytics to develop technologies that deviate substantially from prior art. Overall, these results suggest nuanced deployment in using AI analytics to help mitigate the post-IPO decline in firms that rely on novel

innovation. A possible reason could be that there may be less data for the firm to integrate in creating truly novel innovations (Wu et al. 2020). For the same reasons that innovation declines generally after an IPO, it is also possible that post-IPO firms rely on AI analytics precisely because incentives drive these firms toward more conservative and more combinatorial innovation (Wies and Moorman 2015).

It is possible that our results are driven by selection biases—that firms that go public were poised to experience a decline in innovation prior to their IPO. To address this concern, we follow methods used in the existing literature to address the selection issues around the IPO. Specifically, we adopt PSM method that balances the sample of IPO and non-IPO firms based on the observable characteristics including firm age, total assets (firm size), R&D expenses, patent count and VC financing (Aggarwal and Hsu 2013; Wies and Moorman 2015). The PSM results are consistent (Table E1 in Appendix E). The Heckman selection methods that help rule out the decision to go IPO in the first place also produce consistent results (Tables E2 and E3).

We then adopt a 2SLS estimation approach with the Heckman corrections following our regression specification (Equation 4) to address the potential endogeneity arising from a firm's choice to invest in AI analytics skills (Table E4). The associated F-statistic in the first stage of 2SLS estimation is 16.4, passing the threshold for the weak instrument test, and satisfies the over-identification test (Sargan statistic: $\chi^2 = 2.43, p \leq 0.66$). Finally, we undertake the Heckman estimation procedure in conjunction with instrumental variables on the matched sample produced from the PSM method (Table E5). Overall, these results show that IPOs lower innovation quality, but also that AI analytics can mitigate the decline in the type of innovation that combines existing knowledge yet cannot do so for radically new innovations.

Lastly, if investing in AI analytics is an optimal investment decision for post-IPO firms especially for generating new combinations, we would expect firms to use the resources from their IPO to acquire AI analytics capabilities. Using an event study framework similar to Bernstein (2015), we examine the dynamic changes in AI analytics around an IPO event using Equation 5. Figure 1 shows the changes in AI analytics skills from three years before an IPO to five years afterward. We find that the years following an

IPO are positively associated with an increase in AI analytics capabilities in firms, and the size of the effect becomes larger and more statistically significant over time, peaking at around five years after an IPO event. These results imply that firms are optimally directing resources raised from IPOs to acquire analytics capabilities, potentially to mitigate the post-IPO innovation decline.

Mechanisms

Next, we investigate the main channels by which AI analytics can prevent the post-IPO decline in innovation quality. Three primary reasons for the innovation decline have been identified in the prior literature: (1) the financial market pressure to meet short-term earnings targets, (2) information disclosure, and (3) managerial preference for incremental innovation.

Table 5 shows the results of the mechanism tests. We first test each of the three factors individually. We find that while the short-termism has a negative effect on the new-combination type innovation, AI analytics can be particularly beneficial to those firms with myopic management and can mitigate the negative effects (Column 2). In terms of the information disclosure channel, we examine the time period after 2004, because of the significant changes in information disclosure-related regulations in the US in 2000-2002 (i.e., Regulation Fair Disclosure (Reg FD) in 2000 and the Sarbanes-Oxley SOX) Act in 2002) (Bertomeu et al. 2022). We find that on average the disclosure of information itself has a negative effect on innovation that combines existing technologies in a new way. However, firms having AI analytics capabilities can mitigate the post-IPO innovation loss caused by the disclosure of financial information (with accompanying discussions on risks and firm strategy) to the public (Column 3). We also use an alternative variable about whether a firm is backed by venture capital to capture information disclosure. Results are consistent (Column 4). However, we do not observe the effect of AI analytics in alleviating managerial incentives that may cause post-IPO innovation decline (Column 1). We also estimate the effect of the three factors together (Column 5 and Column 6). We find that AI analytics can help reduce the negative effects of short-termism and information disclosure on innovation, but not of managerial incentives.

Furthermore, we study different industries to examine the heterogeneity in the effects of AI analytics on innovation in these channels. Specifically, information disclosure costs are greater for industries with longer product cycles, such as energy and bio-pharma sectors, but lower for industries with faster product cycles, such as communications equipment and electronic-component sectors (Datar et al. 1997; KPMG 2015). We adopt the industry classification using industry sector codes of the firms (Fama and French 1997; Guo et al. 2004; Kile and Phillips 2009; Popp et al. 2020).²⁰

We observe disclosures to have a negative effect on innovation in energy and bio-pharma sectors perhaps because of their long product cycle. We also find that AI analytics can be particularly helpful in decreasing the negative impact of information disclosure on innovation for the industries with long product cycles. Having AI analytics capability can significantly mitigate the negative effect on innovation due to the high disclosure costs in the two sectors ($\beta_{\ln(\text{Analytics}) \times \text{Disclosure of earnings forecast}} = 0.0551, p < 0.05$ for the energy sector shown in Column 1 of Table 6; $\beta_{\ln(\text{Analytics}) \times \text{Disclosure of earnings forecast}} = 0.0289, p < 0.05$ for the bio-pharma sector shown in Column 3 of Table 6). However, in the sectors with much faster product cycles such as the development of communication and electronics innovation, we find that analytics has a limited effect (close to zero) in mitigating the disclosure effect ($\beta_{\ln(\text{Analytics}) \times \text{Disclosure of earnings forecast}} = 0.000983, p > 0.1$ in Column 5 of Table 6). Using an alternative disclosure measurement with respect to firms receiving venture financing yields similar results (Columns 2, 4, and 6 in Table 6). We also continue to observe similar patterns of results.

Machine Learning Skills, IPO, and Innovation

It is possible that most of the effects we observe with AI analytics comes from machine learning skills. Thus, we conduct exploratory analyses by extending our sample to examine the effect of recent

²⁰ Energy sector includes the industry sector codes NAICS 21, NAICS 22, and NAICS 3336 (Popp et al. 2020). Biotech and pharmaceutical companies are classified in the three-digit Standard Industrial Classification (SIC) code 283 and the four-digit SIC code 8731 (Fama and French 1997; Guo et al. 2004). Communication equipment and electronic components industry sectors include the three-digit SIC code 366 Communication Equipment, and SIC code 367 Electronic Components and Accessories (Kile and Phillips 2009).

advances in artificial intelligence. Here, we only capture skills related to machine learning (Taddy 2019; WIPO 2019). We find that patent originality and new combinations have significantly declined after an IPO, but machine learning skills seem to mitigate the decline to a greater extent than general AI analytics skills that include both ML and other analytics technologies (Table 7). This suggests that the recent advances in machine learning have a greater ability than general AI analytics skills in helping firms find new combinations especially after IPO firms acquired AI analytics skills. Furthermore, we test the role of machine learning skills in countering the three mechanisms through which the innovation quality declines after an IPO. Our findings show that machine learning skills can similarly reduce the pressure from firms' myopic management and meeting information disclosure requirements. However, machine learning skills are limited in its ability to counteract the impact of managerial incentives (Table 8). These results lend support to our theoretical predictions about AI analytics being able to mitigate the decline in innovation for post-IPO firms in these respects.

Lastly, we explore whether the effect on innovation is primarily from innovation in AI technologies. We leverage a dataset of AI-related patents provided by USPTO (Giczy et al. 2022),²¹ and change the dependent variables to number of patents on AI technologies in our estimations. Table 9 presents our results. We find that AI analytics use increases both patents about AI technologies and

²¹ <https://www.uspto.gov/ip-policy/economic-research/research-datasets/artificial-intelligence-patent-dataset>. Giczy et al. (2022) use a machine learning approach to conduct the classification based on the citations and content of the patents including abstracts and claims. They also decompose AI technologies into eight non-exclusive components including machine learning, natural language processing, computer vision, speech, AI hardware, knowledge processing, evolutionary computation, and planning/control. The final output from their predictive model includes eight probability scores between 0 and 1, which indicate the probability of whether a patent includes one of the eight AI components. They define an AI-related patent if one of its eight prediction scores is greater than 0.5. They validate the threshold based on the balance between precision and recall of their model performance, and conclude that 0.5 is a reasonable threshold to determine whether a patent is focused on AI technology or not. Similarly, a patent is machine learning-focused if its probability score of machine learning component is greater than the threshold. Thus, the dataset allows us to explore the impact of AI analytics on AI-focused patents, as well as a specific category of patents related to machine learning in more recent years. A firm's AI-focused innovation is then measured as the total number of AI-related patents in a year, and machine learning-focused innovation can be measured in a similar way. Note that the classified patents available in the dataset were granted till 2020. As we use patent application year to approximate when the actual innovation was produced and started to use in a firm, and the lag between patent application year and grant year is about 2 years (Hall et al. 2001; Lerner and Seru 2022), our analysis is conducted on the sample from 2010 to 2018. Using a sample till the year of 2017 or 2019 yields directionally similar results.

patents about other core technological areas in firms. However, the effect of AI analytics is larger for creating patents in other core technology areas than for creating patents on AI technologies (Column 1 vs. Column 2 for the analysis on the 1988-2013 sample, Column 3 vs. Column 4 for the analysis on the post-2010 sample). This suggests that AI analytics can primarily improve innovation capabilities in the firm's core technology areas as opposed to AI technologies.

We further examine to what extent AI is used in conjunction other information technologies, such as enterprise systems, and the associated impact on innovation that are not just about creating AI technologies. Specifically, as a growing body of literature uses job postings to infer firms' technology exposure, adoption and usage (Acemoglu et al. 2020; Babina et al. 2022; Lou and Wu 2021), we similarly count the number of job postings that require both machine learning skills and enterprise system skills to measure the extent to which AI and enterprise systems are used together. Specifically, we search for a set of skill clusters related to enterprise systems, including Enterprise Resource Planning (ERP), Supply Chain Management (SCM), and Customer Relationship Management (CRM) systems (Hendricks et al. 2007; Hitt et al. 2002). Thus, we separate job posts requiring machine learning skills into (1) those requiring both machine learning and enterprise system skills (*ML and ES*), and (2) those requiring only machine learning skills (*ML-only*).

Our results show that having both machine learning and enterprise system skills has a larger effect on innovation than having just machine learning skills ($\beta_{\ln(ML-only) \times Public} = 0.053, p > 0.1$ in Column 5 vs. $\beta_{\ln(ML \text{ and } ES) \times Public} = 0.136, p < 0.05$ in Column 6; $\beta_{\ln(ML-only) \times Public} = 0.0334, p > 0.1$ vs. $\beta_{\ln(ML \text{ and } ES) \times Public} = 0.148, p < 0.1$ in Column 7). As enterprise systems can facilitate the automation of functional and operational processes and the standardization and integration of a firm's business processes (Jia et al. 2020), firms with these systems have the ability to extract, process and prepare data for downstream applications such as machine learning. The findings also imply that hiring employees with diverse skills, such as the combination of machine learning and enterprise systems can be particularly beneficial for firms whose main innovation does not focus on AI technologies. Furthermore,

to derive benefits from machine learning, our findings suggest that it is critical to have a strong foundation in enterprise systems or related technologies. These systems are critical in enabling firms to prepare and process data so firms can best leverage advanced machine learning tools.

Concluding Remarks

While our study confirms prior work on the post-IPO decline in innovation, our main finding is that AI analytics can mitigate that decline for the type of innovation that combines existing technologies into new ones. Prior work identifies a focus on short-term results, the need for regular public disclosures and a managerial preference for incremental innovation as explanations for the decline. Mitigating this decline has become an important consideration as private firms need to counteract the innovation decline using the cash infusion from an IPO. Firms have tried to address the decline through complex financial instruments such as M&As and LBOs, but these remedies are often very expensive, may provide only temporary relief and can fail. However, AI analytics could provide an alternative and perhaps cheaper way to reverse the innovation decline.

Using a large sample of firms that have gone through an IPO or have remained private as well as measuring their innovative output before and after an IPO, we find AI analytics, as measured by the skillsets of employees, can help post-IPO firms reduce the innovation decline by generating innovations that are new combinations of existing innovations. However, we do not find evidence that AI analytics can help firms create novel innovations that differ substantially from existing ones. Our results are robust to instrumental variables, matching estimators, and Heckman selection models. Furthermore, we also find evidence that firms are increasing their investments in AI analytics post-IPO, suggesting the importance of using AI analytics to shape corporate innovation strategy and allocating resources resulting from IPOs to AI analytics. Our findings also show that AI analytics can stem the innovation decline through some of the channels (short-termism and information disclosure but not the managerial incentives towards incremental innovation), and the effect is stronger for industries with longer product cycles. In exploring how AI helps firms develop innovation, we find that AI analytics primarily helps firms develop

innovation that are not about AI technologies and the effect is stronger when firms have also invested in other information technologies such as enterprise systems.

These results are consistent with earlier findings about the types of innovations that AI analytics can spur and suggest limitations on using AI analytics to address some innovation challenges faced by post-IPO firms. Creating novel innovations that substantially deviate from past inventions remains a challenge for post-IPO firms and has limited remedies. Post-IPO firms should continue to develop strategies to mitigate the decline in novel innovation, perhaps through exploring new technological advances or other organizational incentives. However, for innovation that creates a new combination of existing technologies, AI analytics could be a viable and perhaps superior alternative to other ways of deploying capital to foster innovation.

Our research shows the importance of examining technology as a critical input factor in innovation. While the innovation and entrepreneurship literature has rarely focused on the role of technology in affecting innovation or entrepreneurship outcomes, our work is one of the first to examine the role of AI analytics and related emerging technologies as an important input to innovation outcomes. Similarly, the IT literature has not extensively explored technology investments as a key enabler of innovation. By showing how AI analytics can mitigate the innovation decline post IPO, we bridge both the IT literature and the innovation literature. Our work also differs from other large-scale studies on analytics and related emerging technologies by focusing on private firms especially during their transition to becoming publicly traded.

Overall, our findings support our hypothesis that AI analytics mitigates the innovation decline in post-IPO firms by combining existing technologies into new ones. Furthermore, we do not find evidence that AI analytics can mitigate declines for radically new innovation. While analytics cannot be used to address all forms of innovation decline, future work could explore how advances in artificial intelligence, such as large generative models and potentially other new general-purpose technologies, might further aid innovation, especially for mitigating the innovation decline for post-IPO firms. It is likely that these relationships will continue to evolve as new capabilities in AI analytics are developed (Lou et al. 2023).

Firms with certain capabilities, such as AI analytics, may be especially suited to use these capabilities to innovate and to reduce the post-IPO innovation decline.

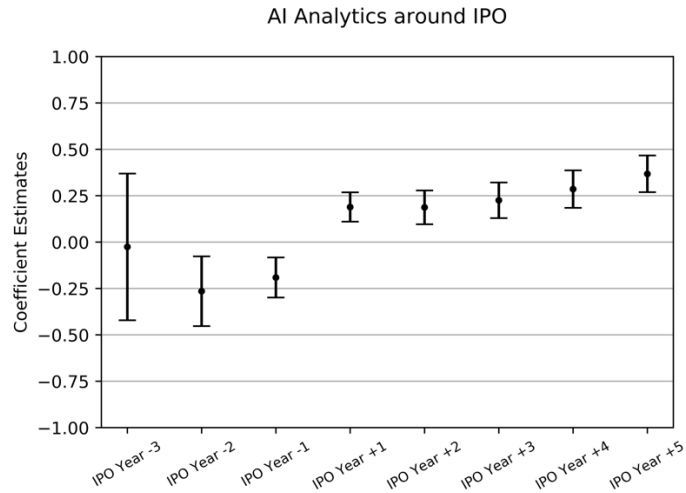


Figure 1. AI Analytics Around the IPO Event

Notes: The unit of analysis is at the firm level. The estimates and confidence intervals (for β_k , where $k = -3, -2, -1, 1, 2, 3, 4, 5$) are taken from the specification: $\ln(Analytics)_{it} = \beta_0 + \sum_{k=-3, k \neq 0}^{k=5} \beta_k IPO Year_{i,k} + y_t + \gamma_i + Controls + \epsilon_{it}$, where $IPO Year_{i,k}$ is a dummy variable that indicates relative k years around IPO. γ_i represents firm fixed effects and y_t indicates year fixed effects. Firm-level controls including firm age, total assets, R&D expenses and total number of patents are also incorporated in the estimation.

Table 1. Summary Statistics and Correlation Table

Variable	Mean	Std dev.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. ln(Originality)	0.92	1.24	1.00															
2. ln(Generality)	0.74	1.11	0.91	1.00														
3. ln(Forward citations, 4 years)	1.93	2.20	0.84	0.90	1.00													
4. ln(Radically new, Global)	0.01	0.12	0.32	0.35	0.25	1.00												
5. ln(Radically new, Local)	0.64	0.90	0.90	0.86	0.78	0.32	1.00											
6. ln(New-combination, Local)	0.86	1.31	0.96	0.89	0.80	0.33	0.83	1.00										
7. ln(Reuse, Local)	0.53	1.03	0.89	0.78	0.68	0.33	0.74	0.88	1.00									
8. Post IPO	0.61	0.49	-0.06	-0.06	-0.02	-0.09	-0.07	-0.07	-0.08	1.00								
9. ln(Analytics)	2.00	2.91	0.13	0.14	0.14	0.06	0.10	0.12	0.10	0.34	1.00							
10. ln(Firm age)	2.59	0.99	0.15	0.08	0.02	0.03	0.12	0.16	0.17	-0.25	-0.09	1.00						
11. ln(Total assets (in million))	5.21	2.33	0.45	0.35	0.26	0.16	0.39	0.46	0.45	-0.08	0.18	0.37	1.00					
12. ln(R&D expense (in million))	2.60	1.69	0.57	0.45	0.39	0.20	0.50	0.58	0.55	-0.01	0.17	0.19	0.69	1.00				
13. ln(Patents)	1.20	1.46	0.98	0.90	0.86	0.31	0.90	0.96	0.89	-0.06	0.12	0.16	0.45	0.58	1.00			
14. Myopic	0.48	0.50	0.18	0.12	0.08	0.02	0.14	0.18	0.19	0.03	0.03	0.24	0.30	0.23	0.18	1.00		
15. VC backed	0.43	0.49	-0.02	0.00	0.06	-0.02	-0.04	-0.03	-0.04	0.09	0.04	-0.16	-0.23	-0.01	-0.01	-0.05	1.00	
16. CEO chaired	0.14	0.35	0.01	0.02	0.03	-0.04	-0.02	0.01	0.02	0.25	0.14	-0.07	-0.01	-0.03	0.01	0.04	0.10	1.00

Notes: We add one to actual values of the variables to avoid the possibilities of taking a natural logarithm of zero.

Table 2. Industry Distribution

Industry by two-digit NAICS	Number of Firms	Percentage of Firms
Accommodation and Food Services	5	0.34%
Administrative and Support and Waste Management and Remediation Services	16	1.09%
Agriculture, Forestry, Fishing and Hunting	5	0.34%
Arts, Entertainment, and Recreation	5	0.34%
Construction	7	0.48%
Health Care and Social Assistance	16	1.09%
Information	273	18.56%
Manufacturing	898	61.05%
Mining, Quarrying, and Oil and Gas Extraction	28	1.90%
Nonclassifiable	23	1.56%
Other Services (except Public Administration)	6	0.41%
Professional, Scientific, and Technical Services	104	7.07%
Real Estate and Rental and Leasing	2	0.14%
Retail Trade	31	2.11%
Transportation and Warehousing	17	1.16%
Utilities	3	0.20%
Wholesale Trade	32	2.18%
Total	1,471	100.00%

Note: The industry classification uses two-digit North American Industry Classification System (NAICS) codes.

Table 3. Firm-level Analyses on Innovation Quality around the IPO Event

Model: FE DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0402*** (0.00872)	-0.0397** (0.0164)	-0.132*** (0.0431)	-0.00223 (0.00349)	0.00147 (0.0151)	-0.0377** (0.0149)	-0.00636 (0.0181)
ln(Firm age)	-0.0142* (0.00858)	0.110*** (0.0268)	0.136*** (0.0473)	0.00981** (0.00468)	0.00273 (0.0136)	0.0148 (0.0129)	-0.0301* (0.0176)
ln(Total assets)	0.00216 (0.00367)	-0.00329 (0.00933)	-0.0361* (0.0200)	0.000799 (0.00132)	-0.00969* (0.00573)	-0.000808 (0.00565)	0.00226 (0.00703)
ln(R&D expense)	0.00922** (0.00441)	-0.0103 (0.0123)	-0.0388 (0.0248)	-0.000729 (0.00188)	-0.0107 (0.00740)	0.0192** (0.00746)	0.0295*** (0.00930)
ln(Patents)	0.793*** (0.00556)	0.663*** (0.00971)	1.525*** (0.0253)	0.0108*** (0.00357)	0.609*** (0.00869)	0.725*** (0.0104)	0.476*** (0.0156)
Observations	14,347	14,347	14,347	14,347	14,347	14,347	14,347
R ²	0.982	0.916	0.887	0.492	0.884	0.957	0.902
Number of firms	1,471	1,471	1,471	1,471	1,471	1,471	1,471
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 4. AI Analytics, Firm Public Status and Innovation Quality

Model: FE DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0596*** (0.00987)	-0.0505*** (0.0181)	-0.143*** (0.0475)	0.00180 (0.00392)	0.00633 (0.0167)	-0.0563*** (0.0174)	-0.0113 (0.0201)
ln(Analytics)	-0.00203 (0.00171)	-0.00205 (0.00432)	-0.0136 (0.0112)	0.00114 (0.000931)	0.00308 (0.00401)	-0.00459 (0.00327)	0.00343 (0.00471)
ln(Analytics) X Post IPO	0.00982*** (0.00191)	0.00558 (0.00449)	0.00741 (0.0115)	-0.00215** (0.000956)	-0.00284 (0.00433)	0.00984*** (0.00343)	0.00194 (0.00524)
ln(Firm age)	-0.0130 (0.00856)	0.111*** (0.0269)	0.137*** (0.0474)	0.00952** (0.00467)	0.00232 (0.0136)	0.0161 (0.0129)	-0.0300* (0.0176)
ln(Total assets)	-0.000225 (0.00371)	-0.00428 (0.00947)	-0.0330* (0.0200)	0.00104 (0.00132)	-0.0100* (0.00585)	-0.00216 (0.00580)	0.000223 (0.00710)
ln(R&D expense)	0.00840* (0.00439)	-0.0107 (0.0123)	-0.0383 (0.0248)	-0.000613 (0.00187)	-0.0107 (0.00741)	0.0186** (0.00748)	0.0289*** (0.00928)
ln(Patents)	0.793*** (0.00555)	0.663*** (0.00971)	1.524*** (0.0253)	0.0108*** (0.00356)	0.610*** (0.00869)	0.725*** (0.0104)	0.476*** (0.0156)
Observations	14,347	14,347	14,347	14,347	14,347	14,347	14,347
R ²	0.982	0.916	0.887	0.492	0.884	0.957	0.902
Number of firms	1,471	1,471	1,471	1,471	1,471	1,471	1,471
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 5. Mechanism Tests

Model: FE DV	(1) ln(New combination, Local)	(2) ln(New combination, Local)	(3) ln(New combination, Local)	(4) ln(New combination, Local)	(5) ln(New combination, Local)	(6) ln(New combination, Local)
ln(Analytics)	0.00466*** (0.00177)	-0.000192 (0.00185)	0.00142 (0.00334)	0.000719 (0.00191)	-0.00551 (0.00430)	-0.00306 (0.00225)
ln(Analytics) X CEO chaired	-0.00323 (0.00328)				-0.00456 (0.00527)	-0.00477 (0.00333)
Myopic		-0.0278*** (0.00976)			-0.0309* (0.0164)	-0.0283*** (0.00974)
ln(Analytics) X Myopic		0.00825*** (0.00224)			0.0120*** (0.00398)	0.00858*** (0.00223)
Disclosure of earnings forecast			-0.0462** (0.0190)		-0.0455** (0.0189)	
ln(Analytics) X Disclosure of earnings forecast			0.0103*** (0.00394)		0.00907** (0.00394)	
ln(Analytics) X VC backed				0.00842*** (0.00312)		0.00932*** (0.00312)
ln(Firm age)	0.00312 (0.0119)	0.00392 (0.0120)	-0.00129 (0.0622)	0.00332 (0.0119)	-0.00587 (0.0625)	0.00377 (0.0119)
ln(Total assets)	-0.00628 (0.00543)	-0.00383 (0.00553)	-0.00251 (0.0122)	-0.00671 (0.00542)	0.000313 (0.0124)	-0.00442 (0.00552)
ln(R&D expense)	0.0191** (0.00742)	0.0171** (0.00744)	0.00423 (0.0123)	0.0188** (0.00742)	0.00201 (0.0126)	0.0167** (0.00742)
ln(Patents)	0.726*** (0.0104)	0.726*** (0.0104)	0.718*** (0.0155)	0.727*** (0.0104)	0.717*** (0.0155)	0.726*** (0.0104)
Observations	14,347	14,347	5,029	14,347	5,029	14,347
R ²	0.957	0.957	0.969	0.957	0.969	0.957
Number of firms	1,471	1,471	931	1,471	931	1,471
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 6. Mechanism Tests by Industries

Model: FE	(1)	(2)	(3)	(4)	(5)	(6)
Industry	Energy	Energy	Bio-pharma	Bio-pharma	Electronics	Electronics
DV	ln(New combination, Local)	ln(New combination, Local)	ln(New combination, Local)	ln(New combination, Local)	ln(New combination, Local)	ln(New combination, Local)
ln(Analytics)	-0.000472 (0.00566)	0.00229 (0.00553)	0.00306 (0.0131)	-0.0104 (0.00876)	-0.00493 (0.0129)	-0.000985 (0.00785)
ln(Analytics) X CEO chaired	-0.00164 (0.00744)	-0.00975 (0.00890)	-0.000102 (0.0199)	-0.0237 (0.0160)	-0.0140 (0.00871)	0.000189 (0.00743)
Myopic	-0.221 (0.242)	-0.230** (0.100)	-0.0385 (0.0373)	-0.00581 (0.0302)	-0.0790* (0.0461)	-0.0503* (0.0278)
ln(Analytics) X Myopic	0.0264 (0.0322)	0.0449** (0.0182)	0.00474 (0.0130)	0.0182** (0.00803)	0.0215* (0.0119)	0.0138** (0.00622)
Disclosure of earnings forecast	-0.325* (0.186)		-0.122* (0.0673)		-0.0413 (0.0613)	
ln(Analytics)X Disclosure of earnings forecast	0.0551** (0.0232)		0.0289** (0.0134)		0.000983 (0.00949)	
ln(Analytics) X VC backed		0.188** (0.0884)		0.0362*** (0.0106)		0.00577 (0.00827)
ln(Firm age)	0.0622 (0.178)	0.0147 (0.0300)	-0.200 (0.154)	0.00540 (0.0388)	-0.0853 (0.186)	-0.0173 (0.0389)
ln(Total assets)	-0.0427 (0.0490)	0.0167 (0.0131)	-0.00815 (0.0327)	-0.000234 (0.0148)	0.00641 (0.0418)	-0.0193 (0.0197)
ln(R&D expense)	-0.0278 (0.0394)	0.0476 (0.0653)	0.0124 (0.0350)	0.0186 (0.0187)	-0.0674 (0.0524)	0.0353 (0.0324)
ln(Patents)	0.680*** (0.0652)	0.643*** (0.0359)	0.586*** (0.0394)	0.659*** (0.0261)	0.788*** (0.0303)	0.775*** (0.0182)
Observations	174	507	810	2,049	618	1,628
R ²	0.955	0.909	0.937	0.942	0.982	0.971
Number of firms	24	37	149	215	113	166
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 7. ML Skills, Firm Public Status and Innovation Quality

Model: FE DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.267** (0.118)	-0.273*** (0.0959)	-0.663** (0.290)	-0.00711 (0.00454)	0.0197 (0.140)	-0.290** (0.128)	-0.190 (0.126)
ln(ML)	-0.0475 (0.0330)	-0.0970 (0.0674)	-0.114 (0.133)	-0.00185 (0.00249)	-0.0520 (0.0577)	-0.00569 (0.0266)	-0.0535 (0.0606)
ln(ML) X Post IPO	0.0771** (0.0380)	0.119 (0.0813)	0.186 (0.161)	0.00290 (0.00327)	0.0959 (0.0782)	0.0845** (0.0409)	-0.00192 (0.0843)
ln(Firm age)	0.0403 (0.0328)	-0.0350 (0.0395)	-0.0453 (0.0793)	0.000117 (0.00513)	-0.0395 (0.0439)	0.114** (0.0489)	-0.0101 (0.0822)
ln(Total assets)	-0.00769 (0.0123)	-0.0196 (0.0167)	0.0207 (0.0323)	0.00157 (0.00195)	-0.00812 (0.0181)	0.00315 (0.0184)	-0.0238 (0.0210)
ln(R&D expense)	0.0351** (0.0170)	0.0230 (0.0181)	0.0112 (0.0347)	0.00270 (0.00245)	0.000476 (0.0222)	-0.0122 (0.0190)	0.0265 (0.0228)
ln(Patents)	0.793*** (0.00910)	0.267*** (0.0155)	0.756*** (0.0184)	0.00943*** (0.00260)	0.581*** (0.0124)	0.745*** (0.0130)	0.646*** (0.0167)
Observations	5,635	5,635	5,635	5,635	5,635	5,635	5,635
R ²	0.974	0.678	0.794	0.272	0.852	0.941	0.908
Number of firms	850	850	850	850	850	850	850
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 8. Mechanism Tests, ML Skills

Model: FE DV	(1) ln(New combination, Local	(2) ln(New combination, Local	(3) ln(New combination, Local	(4) ln(New combination, Local	(5) ln(New combination, Local	(6) ln(New combination, Local
ln(ML)	0.0465 (0.0293)	-0.0574 (0.0570)	-0.00589 (0.0233)	0.0259 (0.0245)	-0.120* (0.0711)	-0.101 (0.0637)
ln(ML) X CEO chaired	0.0280 (0.0584)				0.0362 (0.0623)	0.0275 (0.0478)
Myopic		-0.0154 (0.0178)			-0.0151 (0.0177)	-0.0156 (0.0178)
ln(ML) X Myopic		0.128** (0.0642)			0.122* (0.0637)	0.135** (0.0670)
Disclosure of earnings forecast			-0.0532** (0.0247)		-0.0523** (0.0247)	
ln(ML) X Disclosure of earnings forecast			0.0865** (0.0378)		0.0830* (0.0435)	
ln(ML) X VC backed				0.0990* (0.0599)		0.105* (0.0580)
ln(Firm age)	0.113** (0.0488)	0.115** (0.0489)	0.115** (0.0487)	0.113** (0.0488)	0.117** (0.0488)	0.115** (0.0489)
ln(Total assets)	0.00329 (0.0185)	0.00572 (0.0188)	0.00435 (0.0185)	0.00341 (0.0184)	0.00659 (0.0189)	0.00575 (0.0188)
ln(R&D expense)	-0.0129 (0.0190)	-0.0153 (0.0192)	-0.0129 (0.0190)	-0.0136 (0.0190)	-0.0153 (0.0193)	-0.0161 (0.0193)
ln(Patents)	0.745*** (0.0130)	0.745*** (0.0130)	0.745*** (0.0130)	0.745*** (0.0130)	0.745*** (0.0130)	0.745*** (0.0130)
Observations	5,635	5,635	5,635	5,635	5,635	5,635
R ²	0.941	0.941	0.941	0.941	0.941	0.941
Number of firms	850	850	850	850	850	850
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

Table 9. AI Analytics, Firm Public Status, AI and Non-AI Innovation

Model: FE DV	(1) ln(New combination, AI, Local)	(2) ln(New combination, Non-AI, Local)	(3) ln(New combination, ML, Local)	(4) ln(New combination, Non-ML, Local)	(5) ln(New combination, Non-ML, Local)	(6) ln(New combination, Non-ML, Local)	(7) ln(New combination, Non-ML, Local)
Post IPO	-0.0414** (0.0175)	-0.0438** (0.0180)	-0.0255* (0.0139)	-0.288** (0.128)	-0.288** (0.128)	-0.288** (0.128)	-0.287** (0.128)
ln(Analytics)	-0.000285 (0.00340)	-0.00257 (0.00345)					
ln(Analytics) X Post IPO	0.00618 (0.00392)	0.00717** (0.00363)					
ln(ML)			-0.0270 (0.0164)	-0.00438 (0.0262)			
ln(ML) X Post IPO			0.0408** (0.0206)	0.0829** (0.0406)			
ln(ML-only)					0.00916 (0.0330)		0.0278 (0.0345)
ln(ML-only) X Post IPO					0.0530 (0.0428)		0.0334 (0.0443)
ln(ML and ES)						-0.0336 (0.0299)	-0.0477 (0.0370)
ln(ML and ES) X Post IPO						0.136** (0.0674)	0.148* (0.0763)
ln(Firm age)	-0.0252 (0.0161)	0.0146 (0.0139)	0.0307 (0.0248)	0.112** (0.0489)	0.112** (0.0489)	0.112** (0.0489)	0.112** (0.0489)
ln(Total assets)	0.00297 (0.00628)	-0.00349 (0.00629)	-0.00656 (0.00775)	0.00305 (0.0184)	0.00342 (0.0184)	0.00342 (0.0184)	0.00302 (0.0184)
ln(R&D expense)	0.0235*** (0.00824)	0.0154* (0.00804)	0.00360 (0.00969)	-0.0129 (0.0190)	-0.0130 (0.0190)	-0.0128 (0.0190)	-0.0131 (0.0190)
ln(Patents)	0.206*** (0.0147)	0.672*** (0.0120)	0.0804*** (0.0118)	0.741*** (0.0131)	0.741*** (0.0131)	0.741*** (0.0131)	0.741*** (0.0131)
Observations	14,347	14,347	5,635	5,635	5,635	5,635	5,635
R ²	0.841	0.950	0.645	0.940	0.940	0.940	0.940
Number of firms	1,471	1,471	850	850	850	850	850
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) All models are firm-level fixed effects models.

(2) Robust standard errors are clustered by firm and reported in parentheses.

(3) *** p<0.01, ** p<0.05, * p<0.1

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Appendix A. Measurement of AI Analytics and its Instruments

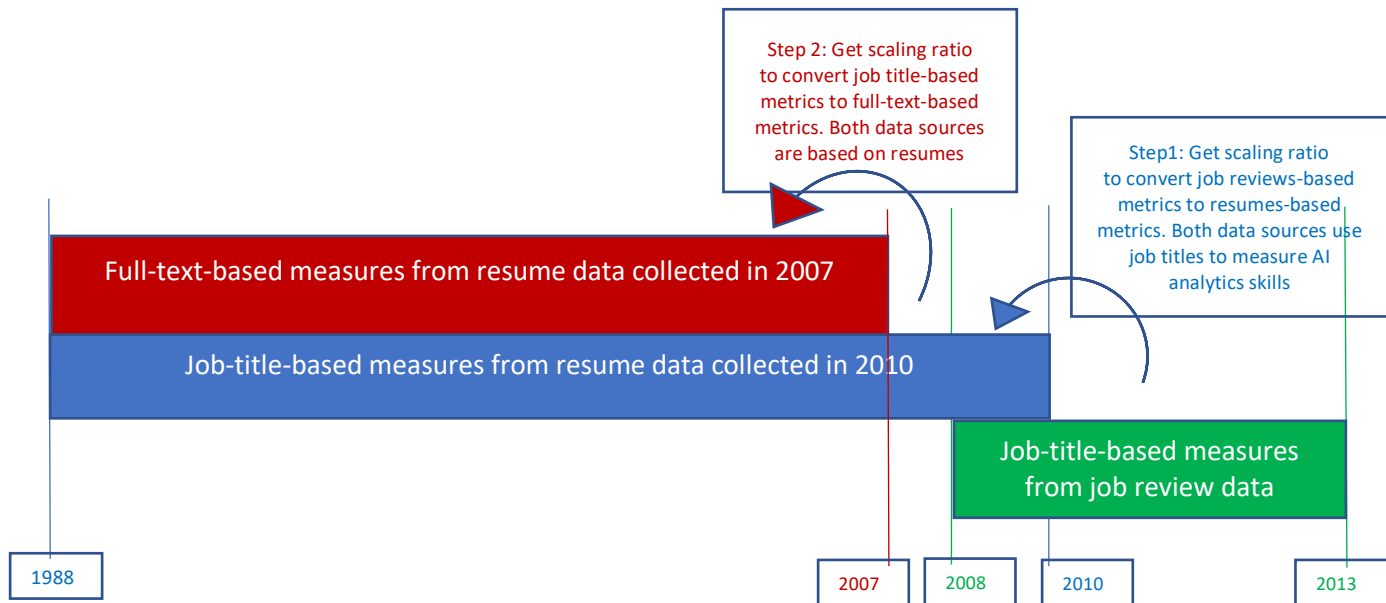


Figure A1: Converting All Metrics to Full-text-resume-based Measures

Notes: Our measurement of AI analytics skills comes from three data sources: 1) resume data with detailed job description from 1988 to 2007; 2) resume data with job titles from 1988 to 2010 (these are updates from the same resume source as (1), so they are consistent); 3) job review data from 2008 to 2013. The boundary years for all the data sources tend to be more incomplete and are thus not good candidates for calculating scaling factors. Thus, to scale source (2) to source (1), we use 2006 as the common year to create scales because 2007 is a boundary year and contains only a subset of the firms in 2006. This serves to translate job-title-based metrics to full-text-based metrics from the same resumes data source. We also translate two different data sources that both use job titles to measure analytics skills (to scale from source (3) to (2)). We primarily use 2009 as the common year to scale because it is not a boundary year. However, when a firm exists in 2008 and 2010 but not in the year 2009, we also use these years to scale. When we cannot find the firm in any of the overlapping years, we scale by using the industry average. This is used in both step 1 and step 2, as shown in Figure A1.

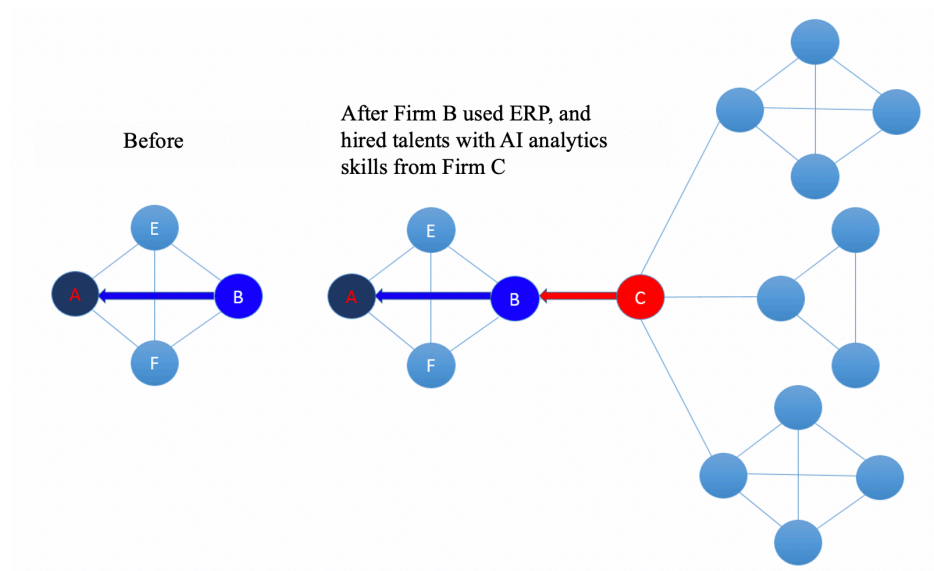


Figure A2: Instrumental Variables

Notes: The directed lines from Firm B to Firm A indicate Firm A can hire employees with AI analytics skills from Firm B. Undirected lines mean that both firms hire from each other. Assume each firm on the left has 10 employees with AI analytics skills. Accordingly, Firm A has access to 30 employees with AI analytics skills. Now Firm B hired 5 more employees with AI analytics skills from Firm C after ERP adoption. Then Firm A accessed to 35 people with AI analytics skills.

Table A1. Taxonomy Used to Classify AI Analytics Skills

General:	Artificial intelligence, Business intelligence, Data analysis, Data center, Data driven, Data fusion, Data integration, Data mining, Data warehouse
System oriented:	Cassandra, Cloud computing, Distributed system, Hadoop, HBase, Google File System, MapReduce
Algorithms & Methodology oriented:	A/B test, Ensemble learning, Genetic algorithm, Machine learning, Natural language processing, Neural network, Network analysis, Optimization, Pattern recognition, Predictive modeling, Regression modeling, Signal processing, Simulation, Supervised learning, Unsupervised learning, Visualization

Notes: We conduct skill-based resume searches and classify AI analytics skills by the taxonomy. The table shows the list of keywords we use that appear frequently in the resume data. We also test several variants of these words, but they don't lead to significant differences for our classifications.

Appendix B: Patent-level Analyses on Innovation Quality around the IPO Event

We replicate Figure 1 in Bernstein (2015) about changes in quality of innovation around the IPO event.

The unit of analysis is at the patent level. The estimates and confidence intervals (for γ_k , where $k = -3, -2, -1, 1, 2, 3, 4, 5$) are taken from the specification: $Y_{it} = \beta_0 + \sum_{k=-3}^{k=5} \gamma_k IPO Year_{i,k} + \tau_i + \mu_t + \epsilon_{i,t}$, where $IPO Year_{i,k}$ is a dummy variable that indicates relative k years around IPO when a patent is applied by firm i in year t . τ_i represents firm fixed effects and μ_t indicates year fixed effects. The dependent variable is originality, generality or all forward citations of successful patent applications by firm i in year t . Firm-level controls including firm age, total assets, R&D expenses and total number of patents are also incorporated in the estimation.

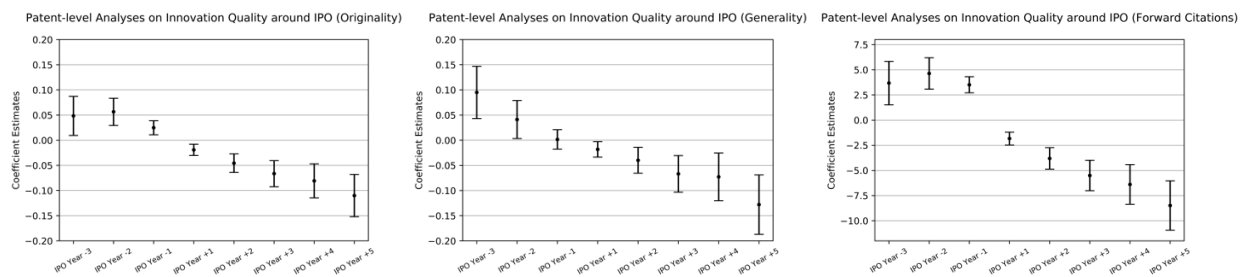


Figure B1. Patent-level Analyses on Innovation Quality around the IPO Event

Appendix C. PSM Procedure

The treatment group in PSM is a set of firms going public and the control group is a set of firms that remain private in our observation period. We use the nearest neighbor matching algorithm to identify the closest $k = 5$ firms in the control group for each treated IPO firms, with a caliper of 0.0001 (Liu and Bharadwaj 2020). The matching is performed based on the observed firm characteristics including firm age, total assets, R&D expense, patent count and VC financing. In Table C1, we examine the covariate balance before and after matching to evaluate the performance of matching. We find that the two groups are statistically different along each observed variable before matching. After matching, the differences of means in the two groups of firms become insignificant across the observed variables, suggesting that selection bias is meaningfully reduced. Figure C1 shows the convergence of the propensity scores distributions for both the treatment and control groups graphically. After matching, we observe more overlap of the propensity scores of the two groups. This suggests that they share more common support on the observed firm characteristics. Our estimation results on the matched sample are shown in Table C2. As robustness checks, we test the matched sample with closest $k = 3$ nearest neighbors, a caliper of 0.001, and we also use Coarsened Exact Matching (CEM) approach (Iacus et al. 2012). Our estimation results on the samples using different matching approaches are directionally consistent (Table C3 for $k = 3$ nearest neighbors, Table C4 for a caliper of 0.001, and Table C5 for CEM).

We also recognize that if there are unobserved variables that simultaneously affect assignment into treatment and the outcome variable, PSM may not be robust against this hidden bias. Thus, we perform Rosenbaum bounds sensitivity analysis (Rosenbaum 2002). Results are shown in Table C6. Parameter Γ is a measure of the degree of departure from no hidden biases, and is used to indicate how strongly an unobserved variable must influence the selection process into the treatment group to undermine the inference of the matching analysis (Khurana et al. 2019). We consider several possible values of Γ . Typically, $\Gamma = 1$ corresponds to the case of the absence of unobserved selection bias, whereas $\Gamma > 2$ offers evidence for robustness to large biases (Proserpio and Zervas 2017). Upper bound and lower bound significance levels show the upper and lower bound for p values of the Wilcoxon sign rank test.

The critical level of Γ at which we would have to question our matching results due to the presence of unobserved variable bias is between 2.2 and 2.3. Our matching results are robust to unobserved selection bias.

Table C1. Differences in Mean Before and After Matching

Variable	Before matching			After matching		
	Treated group	Control group	P-value	Treated group	Control group	P-value
ln(Firm age)	Mean 2.28	Mean 3.24	<0.001	Mean 2.37	Mean 2.37	0.80
ln(Total assets (in million))	Mean 4.89	Mean 5.89	<0.001	Mean 4.92	Mean 4.87	0.22
ln(R&D expense (in million))	Mean 2.50	Mean 2.83	<0.001	Mean 2.51	Mean 2.49	0.69
ln(Patents)	Mean 1.09	Mean 1.44	<0.001	Mean 1.07	Mean 1.04	0.18
ln(VC financing (in million))	Mean 1.11	Mean 0.55	<0.001	Mean 0.93	Mean 0.89	0.17

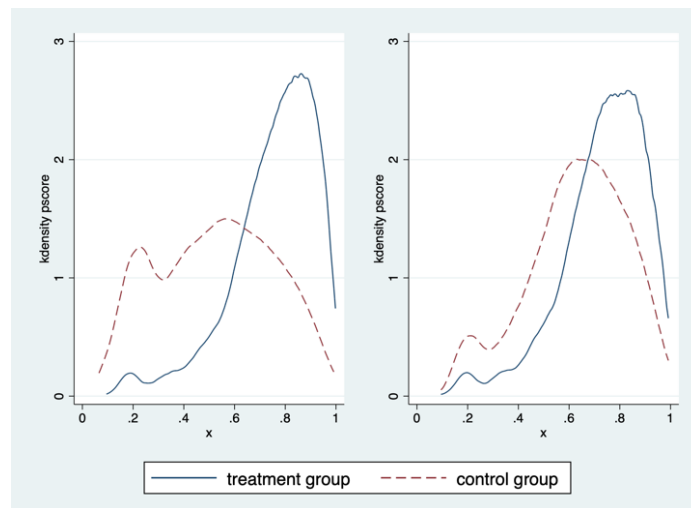


Figure C1. Propensity Score Before and After Matching

Table C2. AI Analytics, Firm Public Status and Innovation Quality

Model: FE/PSM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0638*** (0.0114)	-0.0325* (0.0194)	-0.0949* (0.0574)	0.00333 (0.00440)	-0.0119 (0.0190)	-0.0419** (0.0197)	-0.0153 (0.0227)
ln(Analytics)	-0.00233 (0.00219)	-0.00480 (0.00590)	-0.0178 (0.0145)	0.00115 (0.00104)	-0.00246 (0.00495)	-0.00430 (0.00374)	0.00841 (0.00523)
ln(Analytics) X Post IPO	0.00953*** (0.00240)	0.00871 (0.00606)	0.0195 (0.0149)	-0.00220** (0.00107)	0.00228 (0.00533)	0.00790** (0.00389)	-0.00434 (0.00577)
ln(Firm age)	-0.0164* (0.00936)	0.0961*** (0.0248)	0.215*** (0.0518)	0.0104* (0.00560)	0.0155 (0.0157)	-0.00654 (0.0141)	-0.0552*** (0.0196)
ln(Total assets)	0.000923 (0.00457)	-0.00564 (0.00988)	-0.0326 (0.0224)	0.000602 (0.00123)	-0.00644 (0.00679)	-0.00200 (0.00667)	-0.00231 (0.00869)
ln(R&D expense)	0.00337 (0.00528)	-0.0252** (0.0120)	-0.0514** (0.0252)	8.66×10^{-5} (0.00222)	-0.00804 (0.00855)	0.00870 (0.00867)	0.0261** (0.0108)
ln(Patents)	0.788*** (0.00629)	0.658*** (0.0105)	1.522*** (0.0273)	0.0122*** (0.00414)	0.598*** (0.0104)	0.729*** (0.0109)	0.482*** (0.0163)
Observations	8,977	8,977	8,977	8,977	8,977	8,977	8,977
R ²	0.983	0.924	0.895	0.577	0.891	0.958	0.904
Number of firms	1,426	1,426	1,426	1,426	1,426	1,426	1,426
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses.

(2) *** p<0.01, ** p<0.05, * p<0.1

Table C3. AI Analytics, Firm Public Status and Innovation Quality
(*k* = 3 Nearest Neighbors)

Model: FE/PSM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0636*** (0.0114)	-0.0323* (0.0194)	-0.0972* (0.0574)	0.00303 (0.00438)	-0.0131 (0.0190)	-0.0414** (0.0197)	-0.0143 (0.0227)
ln(Aalytics)	-0.00236 (0.00219)	-0.00478 (0.00590)	-0.0185 (0.0144)	0.00116 (0.00104)	-0.00265 (0.00496)	-0.00426 (0.00375)	0.00882* (0.00523)
ln(Aalytics) X Post IPO	0.00963*** (0.00241)	0.00870 (0.00606)	0.0200 (0.0149)	-0.00218** (0.00107)	0.00246 (0.00534)	0.00791** (0.00390)	-0.00468 (0.00576)
ln(Firm age)	-0.0167* (0.00947)	0.0968*** (0.0248)	0.223*** (0.0518)	0.00991* (0.00547)	0.0139 (0.0155)	-0.00657 (0.0141)	-0.0588*** (0.0195)
ln(Total assets)	0.00121 (0.00460)	-0.00703 (0.00980)	-0.0345 (0.0224)	0.000762 (0.00122)	-0.00624 (0.00675)	-0.00223 (0.00669)	-0.00213 (0.00866)
ln(R&D expense)	0.00258 (0.00539)	-0.0238** (0.0120)	-0.0493* (0.0253)	0.000550 (0.00199)	-0.00748 (0.00847)	0.00809 (0.00866)	0.0266** (0.0105)
ln(Patents)	0.788*** (0.00636)	0.658*** (0.0105)	1.526*** (0.0265)	0.0109*** (0.00307)	0.598*** (0.0105)	0.728*** (0.0106)	0.481*** (0.0160)
Observations	8,927	8,927	8,927	8,927	8,927	8,927	8,927
R ²	0.983	0.924	0.895	0.582	0.892	0.959	0.905
Number of firms	1,425	1,425	1,425	1,425	1,425	1,425	1,425
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses.

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table C4. AI Analytics, Firm Public Status and Innovation Quality
(Caliper of 0.001)**

Model: FE/PSM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0595*** (0.00951)	-0.0415** (0.0167)	-0.154*** (0.0455)	0.00221 (0.00384)	0.00427 (0.0160)	-0.0498*** (0.0168)	-0.00882 (0.0192)
ln(Analytics)	-0.00207 (0.00167)	-0.00110 (0.00427)	-0.0165 (0.0109)	0.00124 (0.000968)	0.00259 (0.00392)	-0.00312 (0.00319)	0.00278 (0.00458)
ln(Analytics) X Post IPO	0.00987*** (0.00186)	0.00500 (0.00438)	0.0105 (0.0111)	-0.00223** (0.000985)	-0.00246 (0.00421)	0.00843** (0.00332)	0.00253 (0.00506)
ln(Firm age)	-0.0125 (0.00817)	0.104*** (0.0249)	0.142*** (0.0465)	0.00936** (0.00451)	-0.00175 (0.0132)	0.0182 (0.0122)	-0.0291* (0.0172)
ln(Total assets)	0.000559 (0.00358)	-0.00529 (0.00883)	-0.0423** (0.0191)	0.000626 (0.00124)	-0.00962* (0.00562)	-0.00173 (0.00560)	0.00161 (0.00681)
ln(R&D expense)	0.00711* (0.00411)	-0.00953 (0.0111)	-0.0344 (0.0230)	-0.000223 (0.00173)	-0.00941 (0.00705)	0.0188*** (0.00722)	0.0301*** (0.00883)
ln(Patents)	0.792*** (0.00530)	0.661*** (0.00908)	1.526*** (0.0236)	0.0108*** (0.00329)	0.609*** (0.00835)	0.721*** (0.00974)	0.478*** (0.0148)
Observations	13,774	13,774	13,774	13,774	13,774	13,774	13,774
R ²	0.981	0.916	0.886	0.495	0.880	0.955	0.896
Number of firms	1,468	1,468	1,468	1,468	1,468	1,468	1,468
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses.

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table C5. AI Analytics, Firm Public Status and Innovation Quality
(Coarsened Exact Matching)**

Model: FE/CEM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0618*** (0.0118)	-0.0625*** (0.0221)	-0.123** (0.0603)	0.00140 (0.00444)	0.00868 (0.0217)	-0.0812*** (0.0214)	-0.0120 (0.0260)
ln(Analytics)	-0.00319 (0.00246)	-0.00528 (0.00669)	-0.0143 (0.0170)	-9.58×10 ⁻⁵ (0.00124)	0.00475 (0.00582)	-0.00920* (0.00479)	0.00185 (0.00720)
ln(Analytics) X Post IPO	0.0103*** (0.00265)	0.00996 (0.00667)	0.00677 (0.0169)	-0.00126 (0.00125)	-0.00569 (0.00612)	0.0138*** (0.00492)	0.00313 (0.00748)
ln(Firm age)	-0.00273 (0.0108)	0.0687** (0.0269)	0.159*** (0.0602)	0.00492 (0.00437)	-0.0129 (0.0164)	0.00429 (0.0155)	-0.0385* (0.0222)
ln(Total assets)	-0.00340 (0.00570)	0.0178 (0.0113)	0.00687 (0.0281)	0.000381 (0.00154)	-0.00356 (0.00833)	-0.00637 (0.00765)	-0.00825 (0.0103)
ln(R&D expense)	0.0115* (0.00614)	-0.0274* (0.0150)	-0.0817** (0.0348)	-0.000975 (0.00179)	-0.00673 (0.0101)	0.0117 (0.00971)	0.0356*** (0.0135)
ln(Patents)	0.784*** (0.00741)	0.656*** (0.0107)	1.558*** (0.0269)	0.00669*** (0.00133)	0.596*** (0.0110)	0.708*** (0.0110)	0.456*** (0.0189)
Observations	8,732	8,732	8,732	8,732	8,732	8,732	8,732
R ²	0.979	0.918	0.889	0.496	0.878	0.952	0.889
Number of firms	1,242	1,242	1,242	1,242	1,242	1,242	1,242
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses.

(2) *** p<0.01, ** p<0.05, * p<0.1

Table C6. Rosenbaum Bounds Test Results

Γ	Lower bound significance level	Upper bound significance level
1.0	0	0
1.1	0	0
1.2	0	0
1.3	0	0
1.4	0	0
1.5	0	0
1.6	0	0
1.7	0	0
1.8	0	0
1.9	0	1.1×10^{-16}
2.0	0	2.2×10^{-10}
2.1	0	7.6×10^{-6}
2.2	0	0.0062
2.3	0	0.22
2.4	0	0.82
2.5	0	0.99
2.6	0	1
2.7	0	1
2.8	0	1
2.9	0	1
3.0	0	1

Appendix D. ML Skills

Table D1. A List of Exemplar Keywords for Identifying ML Skills

Decision Tree	Keras	Machine Learning	MXNet	Semi-Supervised Learning	Torch
Deeplearning4j	Latent Dirichlet Allocation	Madlib	Neural Network	Supervised Learning	Unsupervised Learning
Deep Learning	Libsvm	Mahout	Pybrain	Support Vector Machines (SVM)	Word2Vec
Gradient boosting	Long Short-Term Memory (LSTM)	Mlpy	Random Forests	TensorFlow	Xgboost

Appendix E: Matching Estimators and Heckman Selection

Table E1. AI Analytics, Firm Public Status and Innovation Quality

Model: FE/PSM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0638*** (0.0114)	-0.0325* (0.0194)	-0.0949* (0.0574)	0.00333 (0.00440)	-0.0119 (0.0190)	-0.0419** (0.0197)	-0.0153 (0.0227)
ln(Analytics)	-0.00233 (0.00219)	-0.00480 (0.00590)	-0.0178 (0.0145)	0.00115 (0.00104)	-0.00246 (0.00495)	-0.00430 (0.00374)	0.00841 (0.00523)
ln(Analytics) X Post IPO	0.00953*** (0.00240)	0.00871 (0.00606)	0.0195 (0.0149)	-0.00220** (0.00107)	0.00228 (0.00533)	0.00790** (0.00389)	-0.00434 (0.00577)
ln(Firm age)	-0.0164* (0.00936)	0.0961*** (0.0248)	0.215*** (0.0518)	0.0104* (0.00560)	0.0155 (0.0157)	-0.00654 (0.0141)	-0.0552*** (0.0196)
ln(Total assets)	0.000923 (0.00457)	-0.00564 (0.00988)	-0.0326 (0.0224)	0.000602 (0.00123)	-0.00644 (0.00679)	-0.00200 (0.00667)	-0.00231 (0.00869)
ln(R&D expense)	0.00337 (0.00528)	-0.0252** (0.0120)	-0.0514** (0.0252)	8.66×10^{-5} (0.00222)	-0.00804 (0.00855)	0.00870 (0.00867)	0.0261** (0.0108)
ln(Patents)	0.788*** (0.00629)	0.658*** (0.0105)	1.522*** (0.0273)	0.0122*** (0.00414)	0.598*** (0.0104)	0.729*** (0.0109)	0.482*** (0.0163)
Observations	8,977	8,977	8,977	8,977	8,977	8,977	8,977
R ²	0.983	0.924	0.895	0.577	0.891	0.958	0.904
Number of firms	1,426	1,426	1,426	1,426	1,426	1,426	1,426
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses.

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table E2. AI Analytics, Firm Public Status and Innovation Quality
(Heckman Corrections)**

Model: Heckman DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0589*** (0.00620)	-0.0553*** (0.0203)	-0.152*** (0.0168)	0.00184 (0.00462)	0.00545 (0.0286)	-0.0558*** (0.0183)	-0.00712 (0.0176)
ln(Analytics)	-0.00194 (0.00236)	-0.00269 (0.00562)	-0.0148 (0.0114)	0.00115 (0.000835)	0.00296 (0.00319)	-0.00452 (0.00306)	0.00399 (0.00465)
ln(Analytics) X Post IPO	0.00971*** (0.00274)	0.00629 (0.00385)	0.00877 (0.00974)	-0.00215** (0.00106)	-0.00271 (0.00341)	0.00977*** (0.00324)	0.00132 (0.00507)
ln(Firm age)	-0.0236 (0.0144)	0.180*** (0.0650)	0.271* (0.140)	0.00891 (0.00601)	0.0149 (0.0162)	0.00914 (0.0125)	-0.0903 (0.0571)
ln(Total assets)	0.000348 (0.00406)	-0.00802 (0.00700)	-0.0402** (0.0171)	0.00107 (0.00147)	-0.0107 (0.00890)	-0.00178 (0.00213)	0.00350 (0.00435)
ln(R&D expense)	0.0111** (0.00524)	-0.0286 (0.0197)	-0.0730 (0.0499)	-0.000455 (0.00369)	-0.0139 (0.0105)	0.0205*** (0.00646)	0.0446*** (0.0151)
ln(Patents)	0.790*** (0.00555)	0.681*** (0.00950)	1.560*** (0.0429)	0.0107* (0.00552)	0.613*** (0.00856)	0.723*** (0.00420)	0.460*** (0.0143)
IMR	0.0866 (0.102)	-0.565 (0.375)	-1.094 (0.937)	0.00501 (0.0452)	-0.103 (0.0975)	0.0572 (0.111)	0.494 (0.399)
Observations	14,347	14,347	14,347	14,347	14,347	14,347	14,347
Number of firms	1,471	1,471	1,471	1,471	1,471	1,471	1,471
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses. Bootstrapped standard errors are calculated as suggested in the literature (e.g., Wies and Moorman 2015; Xiong and Bharadwaj 2011).

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table E3. AI Analytics, Firm Public Status and Innovation Quality
(Heckman Corrections with PSM)**

Model: Heckman with PSM DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0638*** (0.0190)	-0.0324** (0.0165)	-0.0949* (0.0506)	0.00333 (0.00733)	-0.0119 (0.0153)	-0.0419** (0.0205)	-0.0153 (0.0293)
ln(Analytics)	-0.00232 (0.00238)	-0.00493 (0.00711)	-0.0179 (0.0164)	0.00116 (0.00122)	-0.00253 (0.00726)	-0.00430 (0.00386)	0.00851 (0.00555)
ln(Analytics) X Post IPO	0.00949*** (0.00240)	0.00905 (0.00675)	0.0199 (0.0132)	-0.00221** (0.000961)	0.00244 (0.00666)	0.00788*** (0.00245)	-0.00457 (0.00464)
ln(Firm age)	-0.0226 (0.0197)	0.156*** (0.0400)	0.289*** (0.0675)	0.00864** (0.00373)	0.0434 (0.0391)	-0.00964 (0.0134)	-0.0965** (0.0450)
ln(Total assets)	0.00286 (0.00440)	-0.0241 (0.0154)	-0.0557 (0.0437)	0.00114 (0.00293)	-0.0151 (0.0116)	-0.00103 (0.00554)	0.0106 (0.00885)
ln(R&D expense)	0.00506 (0.00558)	-0.0414 (0.0316)	-0.0715** (0.0333)	0.000553 (0.00191)	-0.0156 (0.0122)	0.00954 (0.00940)	0.0374** (0.0151)
ln(Patents)	0.787*** (0.00543)	0.669*** (0.00389)	1.535*** (0.0350)	0.0119*** (0.00367)	0.603*** (0.0110)	0.729*** (0.00622)	0.475*** (0.0140)
IMR	0.126 (0.192)	-1.205 (0.804)	-1.500 (1.154)	0.0349 (0.114)	-0.567 (0.462)	0.0629 (0.195)	0.839 (0.675)
Observations	8,977	8,977	8,977	8,977	8,977	8,977	8,977
Number of firms	1,426	1,426	1,426	1,426	1,426	1,426	1,426
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses. Bootstrapped standard errors are calculated as suggested in the literature (e.g., Wies and Moorman 2015; Xiong and Bharadwaj 2011).

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table E4. AI Analytics, Firm Public Status and Innovation Quality
(Heckman Corrections with 2SLS)**

Model: Heckman with 2SLS DV	(1) ln(Originality)	(2) ln(Generality)	(3) ln(Forward citations, 4 years)	(4) ln(Radically new, Global)	(5) ln(Radically new, Local)	(6) ln(New combination, Local)	(7) ln(Reuse, Local)
Post IPO	-0.0914*** (0.0352)	-0.261** (0.130)	-0.209** (0.104)	0.0929 (0.0587)	0.0238 (0.174)	-0.111*** (0.0192)	0.441 (0.757)
ln(Analytics)	-0.0424 (0.0268)	-0.167 (0.144)	-0.329 (0.311)	0.0858 (0.0719)	-0.0519 (0.174)	0.0173 (0.0672)	0.518 (0.570)
ln(Analytics) X Post IPO	0.0299** (0.0149)	0.132 (0.0837)	0.0863 (0.102)	-0.0589* (0.0345)	-0.000345 (0.116)	0.0324** (0.0157)	-0.290 (0.419)
ln(Firm age)	0.00860 (0.0192)	0.0838*** (0.0257)	0.194 (0.123)	-0.0314 (0.0396)	0.0203 (0.0441)	-0.00364 (0.0276)	-0.140 (0.148)
ln(Total assets)	0.0105 (0.00918)	0.0287 (0.0362)	0.0665 (0.100)	-0.0136 (0.0194)	0.0107 (0.0328)	-0.0163 (0.0267)	-0.115 (0.118)
ln(R&D expense)	0.0126 (0.00965)	0.0138 (0.0205)	0.0296 (0.0363)	-0.00225 (0.00438)	-0.00332 (0.0183)	0.0205 (0.0205)	-0.0260 (0.0499)
ln(Patents)	0.791*** (0.00739)	0.705*** (0.0107)	1.661*** (0.0502)	0.00577*** (0.00194)	0.624*** (0.00692)	0.707*** (0.00872)	0.465*** (0.0314)
IMR	-0.114 (0.0930)	0.0456 (0.123)	-0.0372 (0.305)	0.133 (0.0857)	-0.0454 (0.0870)	0.00648 (0.105)	-0.158 (0.119)
Observations	11,792	11,792	11,792	11,792	11,792	11,792	11,792
Number of firms	1,388	1,388	1,388	1,388	1,388	1,388	1,388
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses. Bootstrapped standard errors are calculated as suggested in the literature (e.g., Wies and Moorman 2015; Xiong and Bharadwaj 2011).

(2) *** p<0.01, ** p<0.05, * p<0.1

**Table E5. AI Analytics, Firm Public Status and Innovation Quality
(Heckman Corrections with 2SLS and PSM)**

Model: Heckman with 2SLS and PSM DV	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ln(Originality)	ln(Generality)	ln(Forward citations, 4 years)	ln(Radically new, Global)	ln(Radically new, Local)	ln(New combination, Local)	ln(Reuse, Local)
Post IPO	-0.0755*** (0.0210)	-0.0690** (0.0345)	-0.585* (0.344)	0.0362 (0.0220)	-0.0103 (0.107)	-0.0922*** (0.0122)	0.144 (0.183)
ln(Analytics)	0.00659 (0.00807)	-0.0537 (0.0408)	-0.589 (0.373)	0.0194 (0.0163)	-0.0879 (0.111)	0.0140 (0.0605)	0.281 (0.172)
ln(Analytics) X Post IPO	0.0118** (0.00479)	0.0364 (0.0276)	0.378 (0.257)	-0.0224** (0.0112)	0.0238 (0.0782)	0.0256** (0.0104)	-0.145 (0.122)
ln(Firm age)	-0.0182 (0.0130)	0.0468* (0.0258)	0.258* (0.133)	0.00351 (0.00752)	0.0244 (0.0225)	-0.0210 (0.0159)	-0.0916 (0.0707)
ln(Total assets)	-0.00704 (0.00858)	0.00666 (0.00780)	0.0400 (0.0590)	0.00224 (0.00316)	0.0141 (0.0215)	-0.0130 (0.0223)	-0.0441 (0.0308)
ln(R&D expense)	0.00902 (0.00584)	-0.00223 (0.0198)	0.0226 (0.0166)	0.00183 (0.00170)	-0.000916 (0.0126)	0.0165 (0.0238)	-0.00273 (0.0153)
ln(Patents)	0.781*** (0.00580)	0.708*** (0.0102)	1.671*** (0.0269)	0.00930*** (0.00199)	0.621*** (0.00468)	0.709*** (0.0114)	0.444*** (0.0116)
IMR	-0.0810 (0.0509)	0.0677 (0.0838)	0.146 (0.280)	0.0216 (0.0158)	0.0287 (0.0521)	0.0603 (0.151)	-0.0663 (0.163)
Observations	7,241	7,241	7,241	7,241	7,241	7,241	7,241
Number of firms	1,200	1,200	1,200	1,200	1,200	1,200	1,200
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES	YES	YES

Notes:

(1) Robust standard errors are clustered by firm and reported in parentheses. Bootstrapped standard errors are calculated as suggested in the literature (e.g., Wies and Moorman 2015; Xiong and Bharadwaj 2011).

(2) *** p<0.01, ** p<0.05, * p<0.1